Lab 03: Working with Git History, Stashing, and Reverting Commits

Name: Fatima Kalsoom

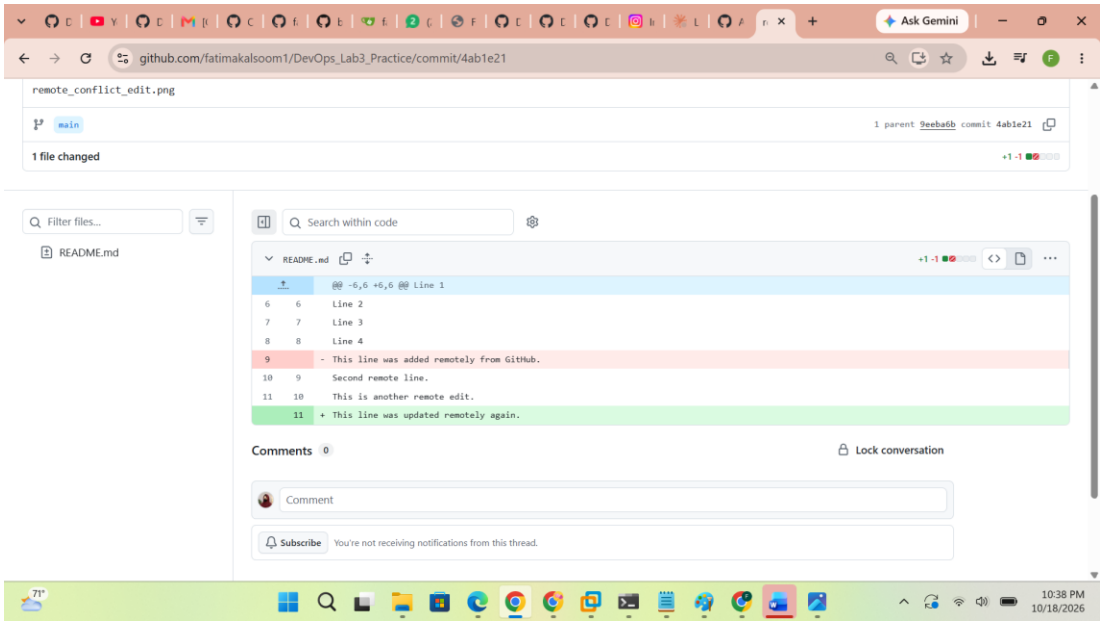
Registration Number: 2023-BSCS-32

Course Code: CS-649

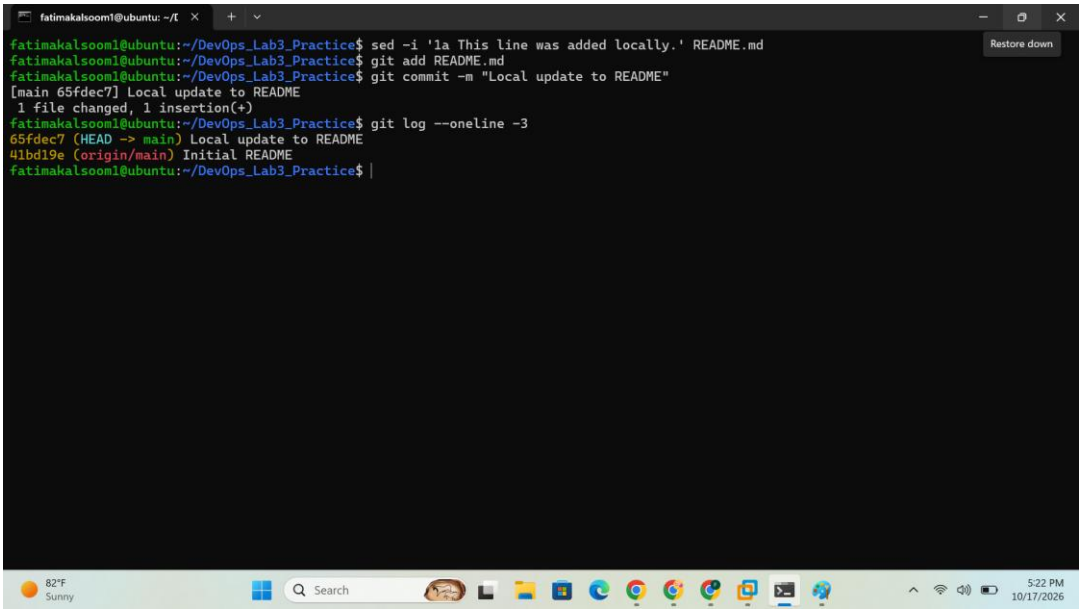
Section: A

GitHub Username: fatimakalsoom1

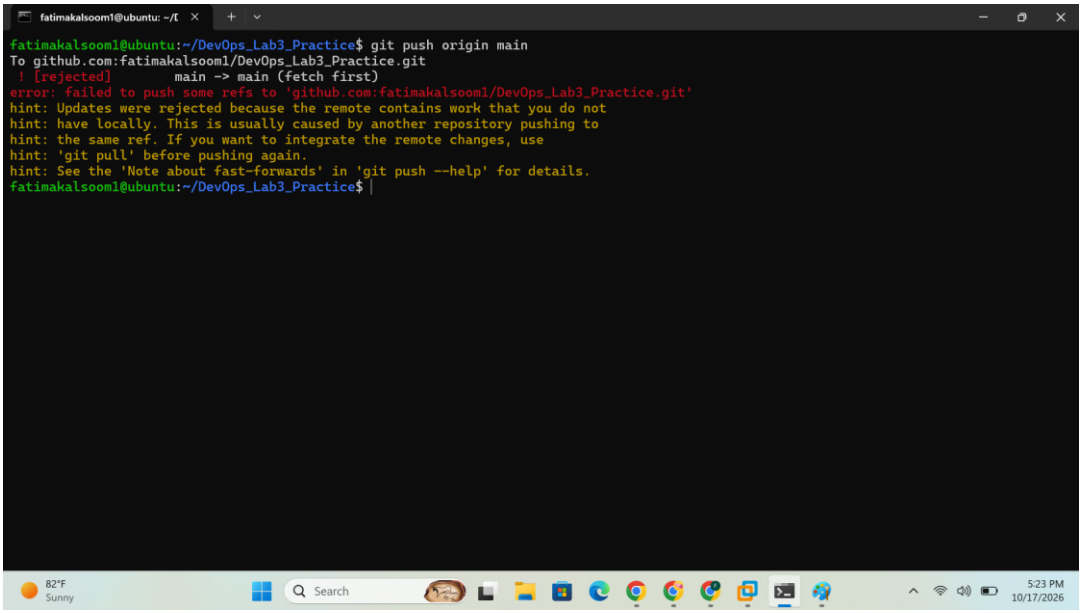
Task 1



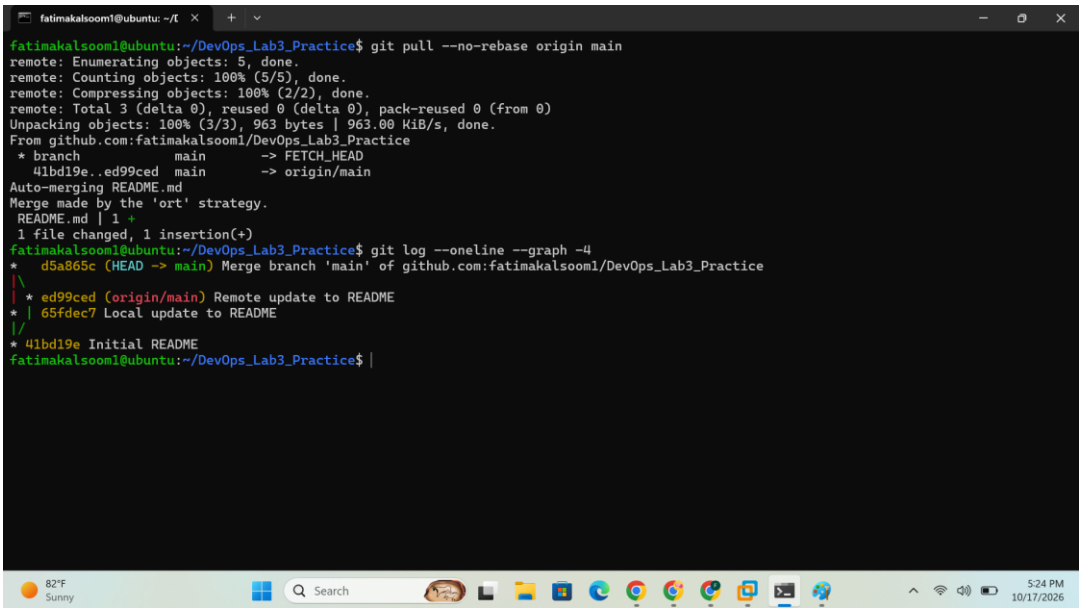
Remote edit made on GitHub: a new line added to README.md and committed (commit 658e876).



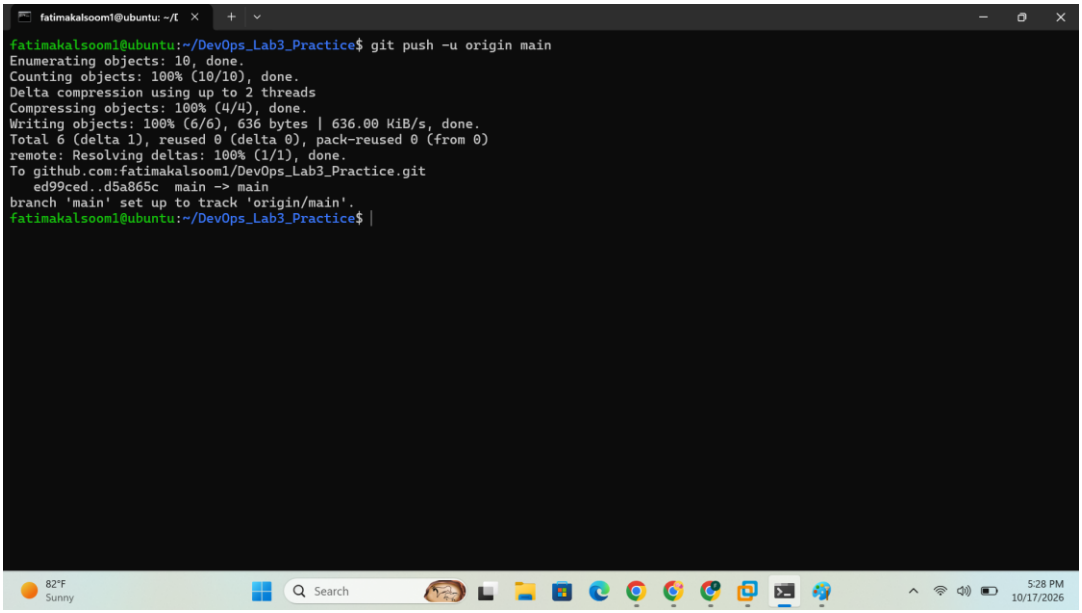
Local update to README committed and verified with git log.



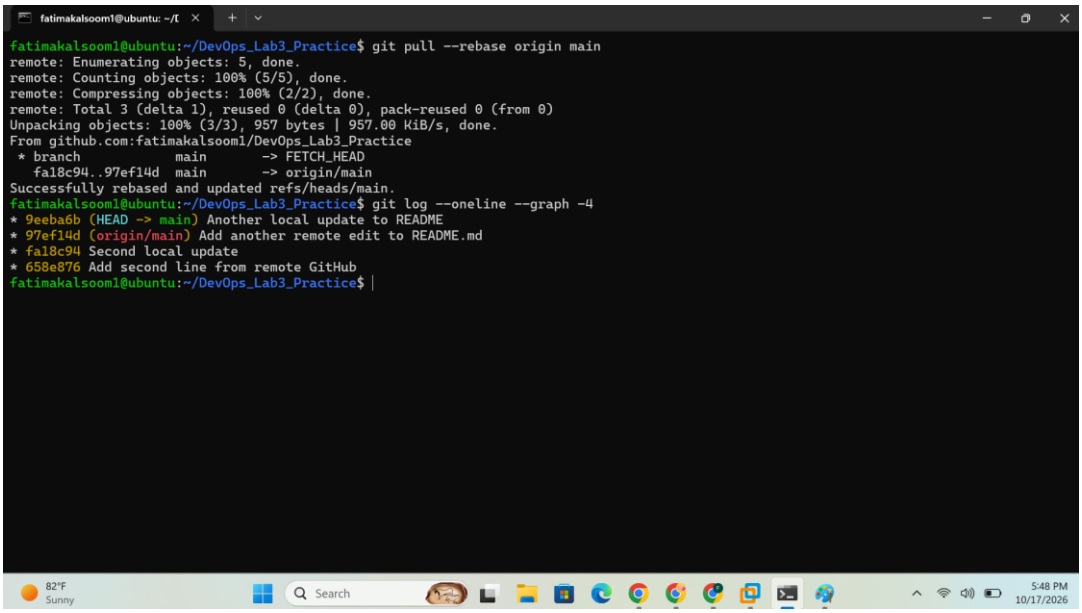
Push rejected because the remote contains work that is not present locally.



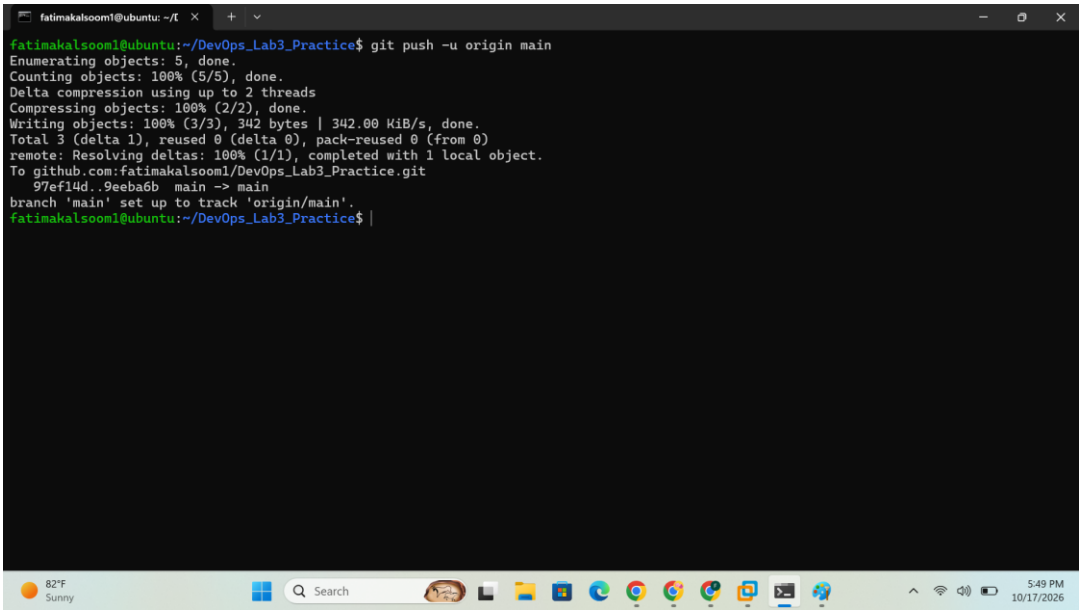
git pull --no-rebase merges the remote changes and creates a merge commit.



Merged history pushed successfully to origin/main.

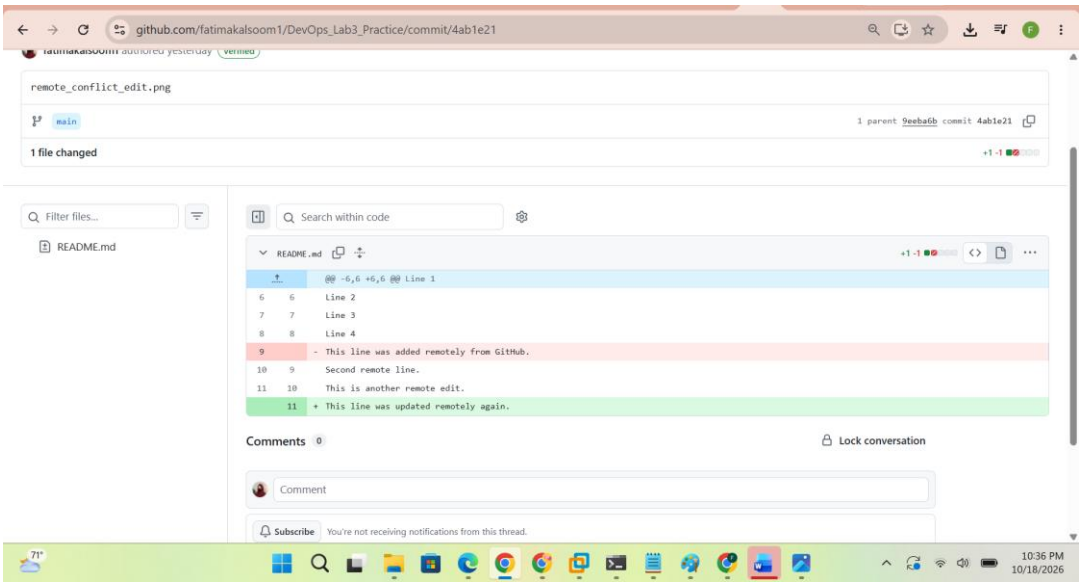


git pull --rebase replays the local commit on top of the pulled remote commits.

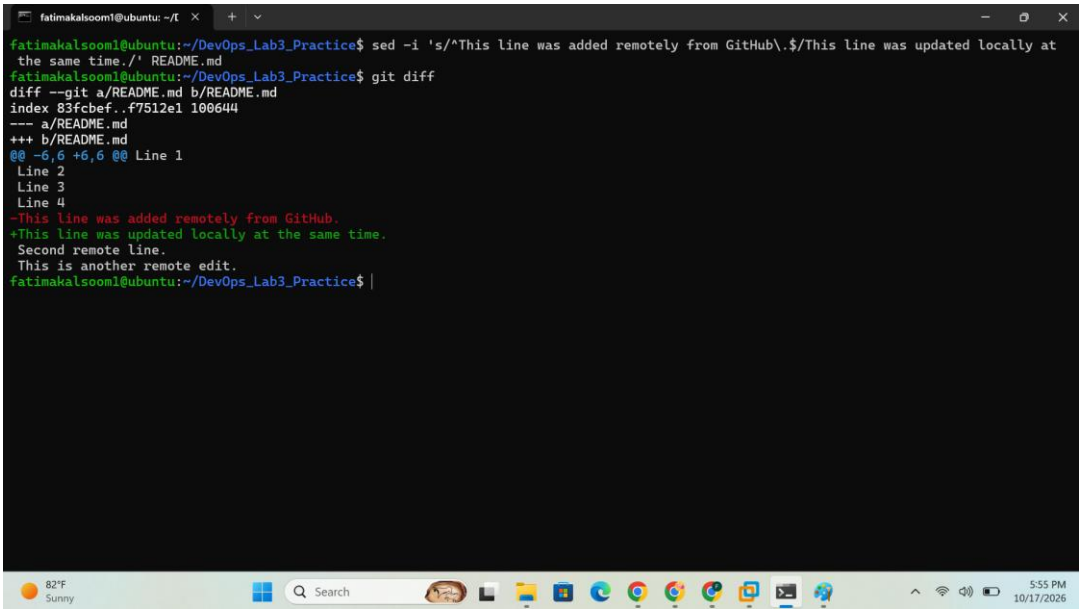


Rebased commits pushed successfully to origin/main.

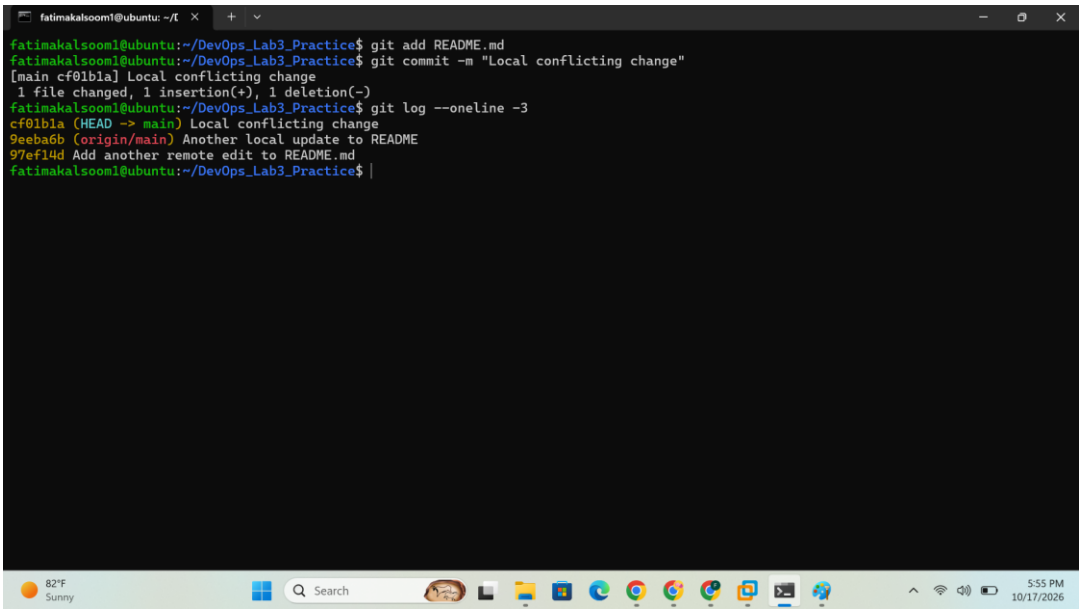
Task 2



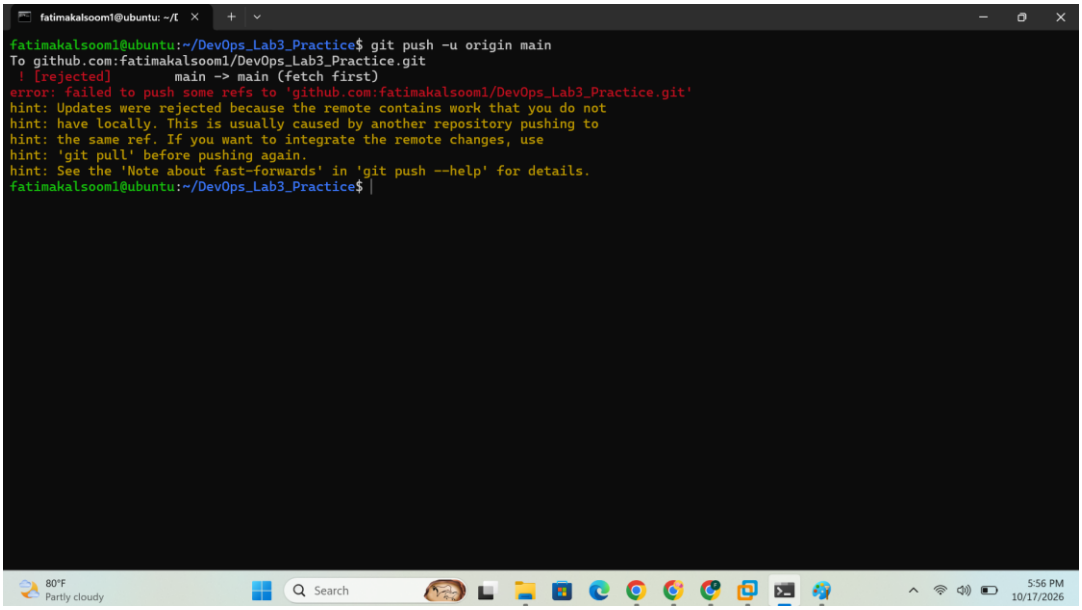
Conflicting change committed on GitHub: the remote line was changed to "This line was updated remotely again."



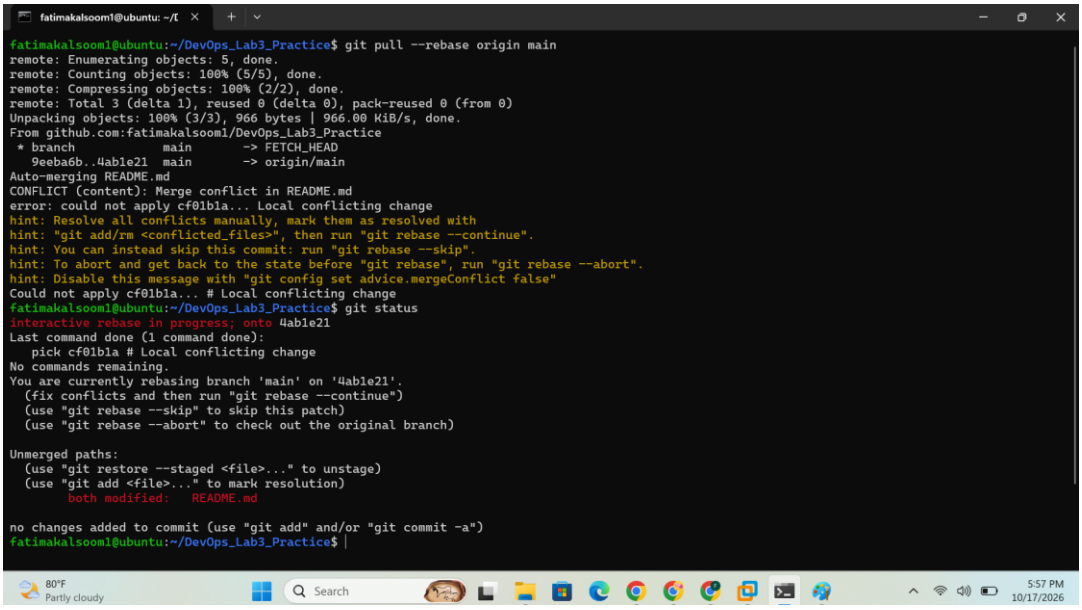
Same README.md line edited locally; git diff shows the local change.



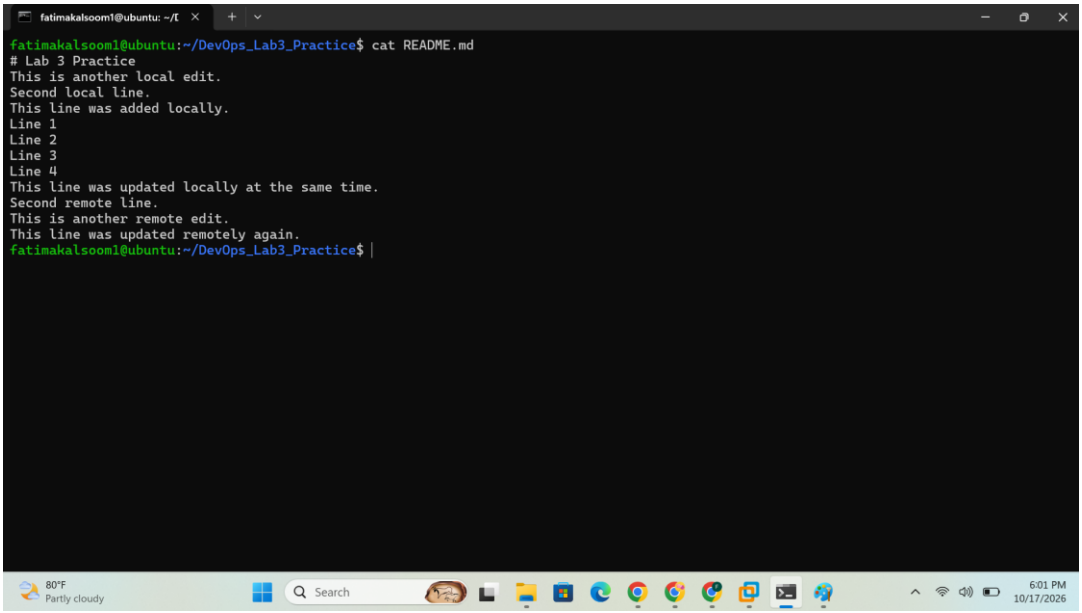
Conflicting local change committed.



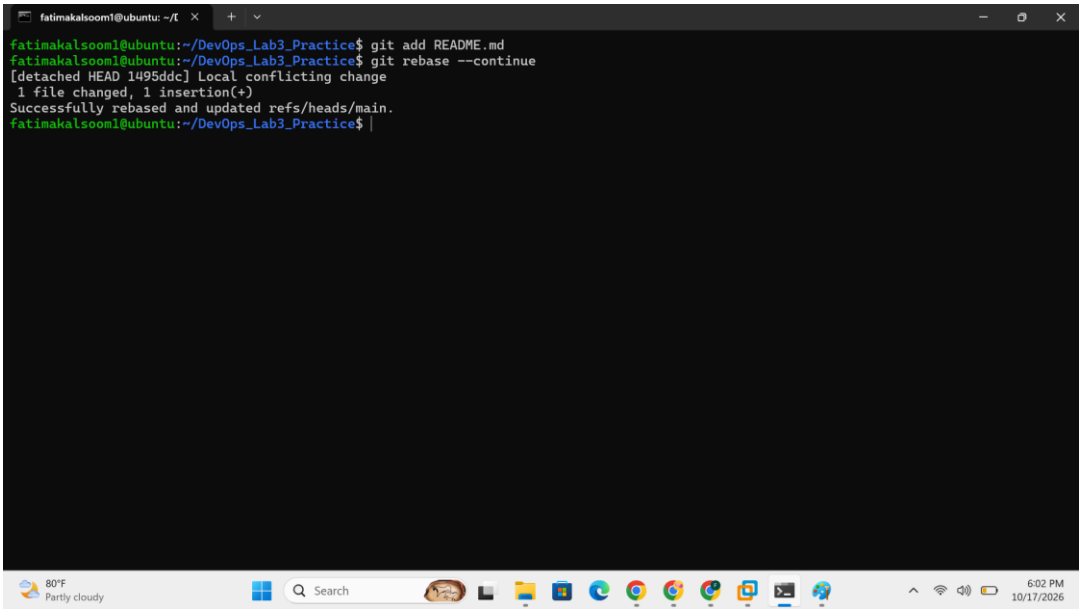
Push rejected because the remote has conflicting changes.



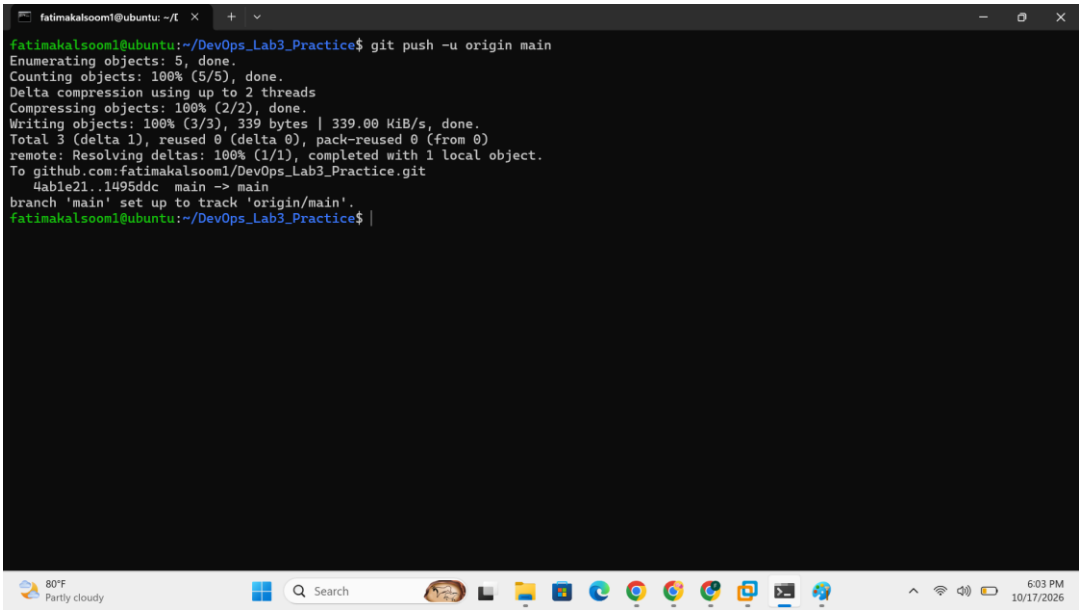
Rebase paused with a conflict in README.md; git status lists the unmerged path.



README.md after manually resolving the conflict markers.

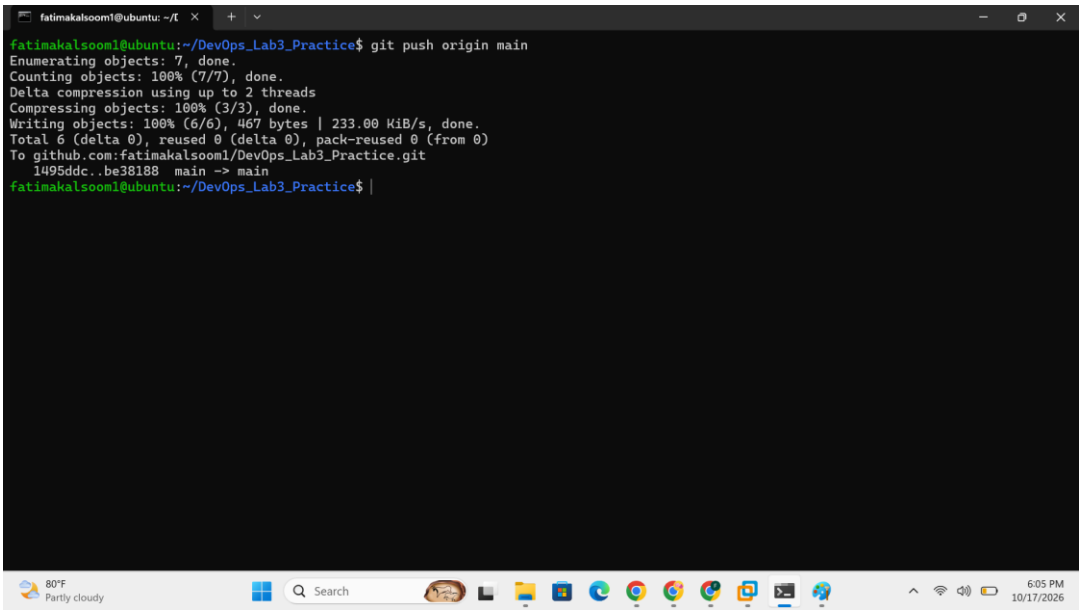


git add and git rebase --continue complete the rebase.

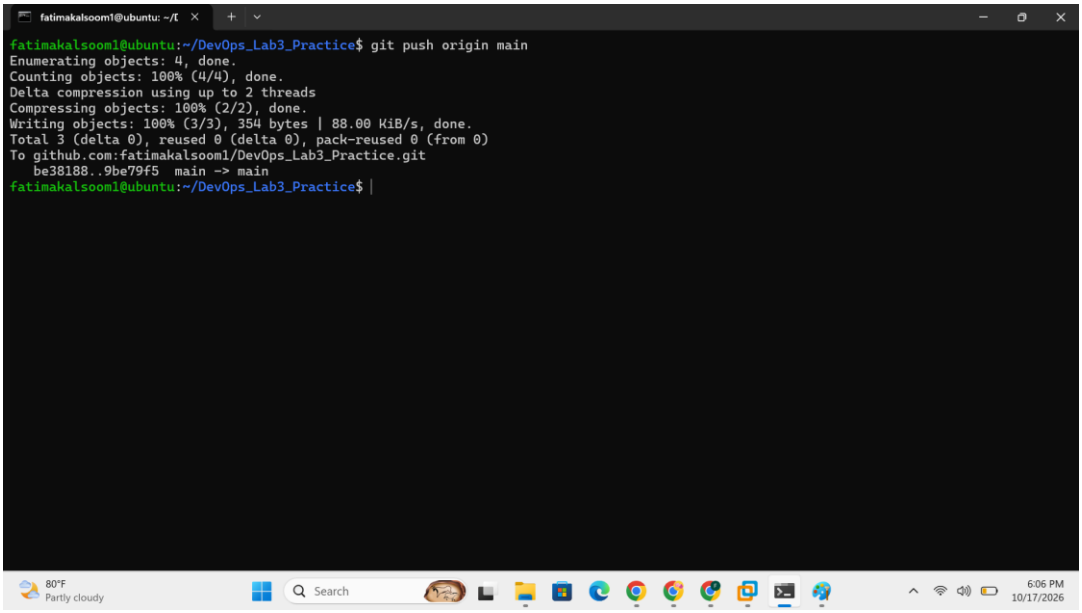


Resolved changes pushed successfully to origin/main.

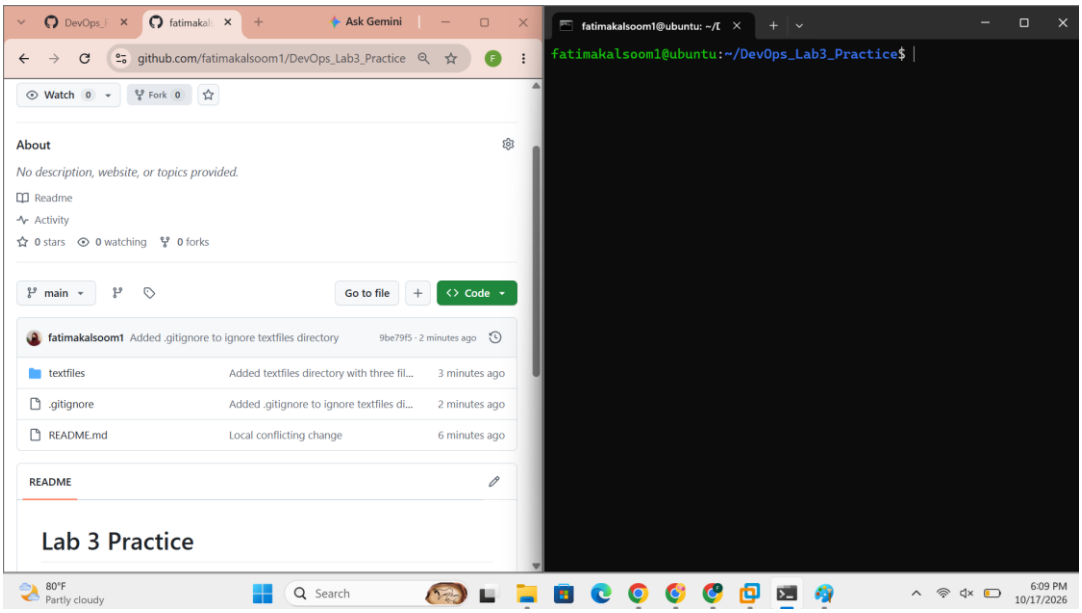
Task 3



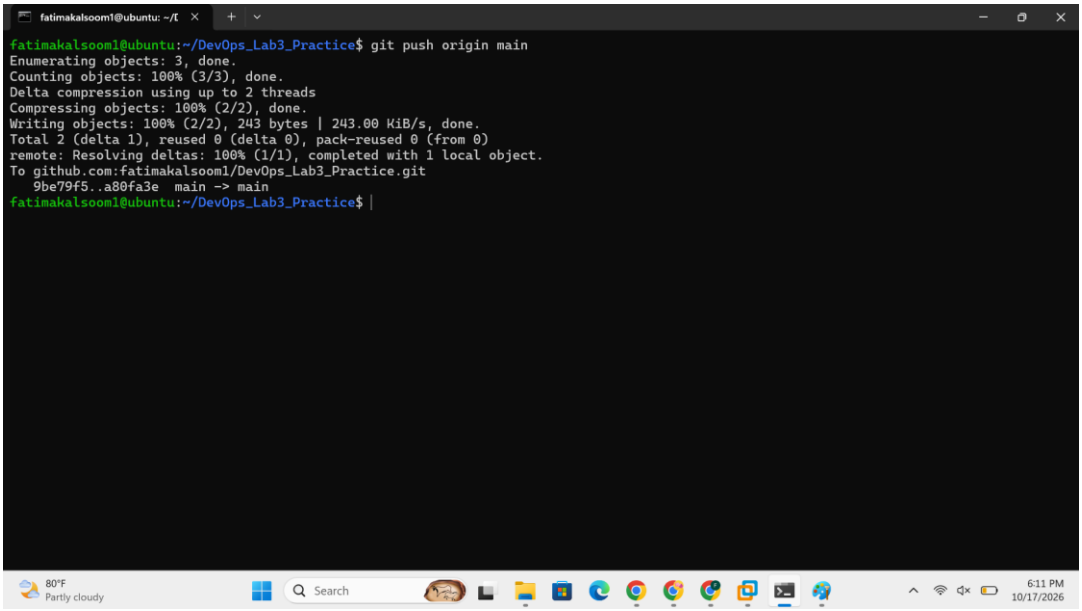
textfiles directory with three files committed and pushed.



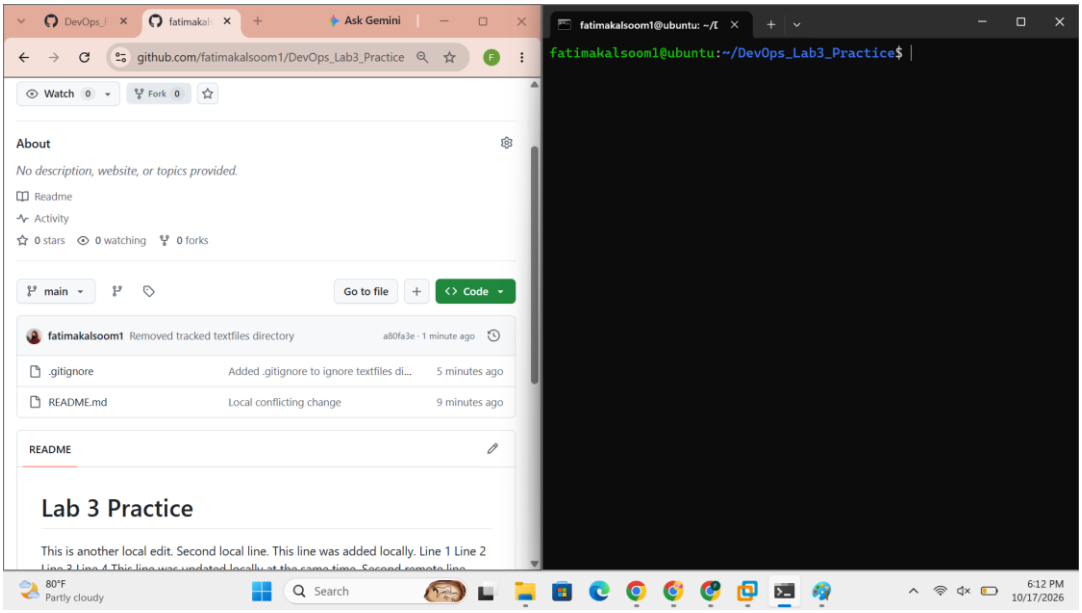
.gitignore added to ignore the textfiles directory and pushed.



GitHub still shows textfiles because the files were already tracked.

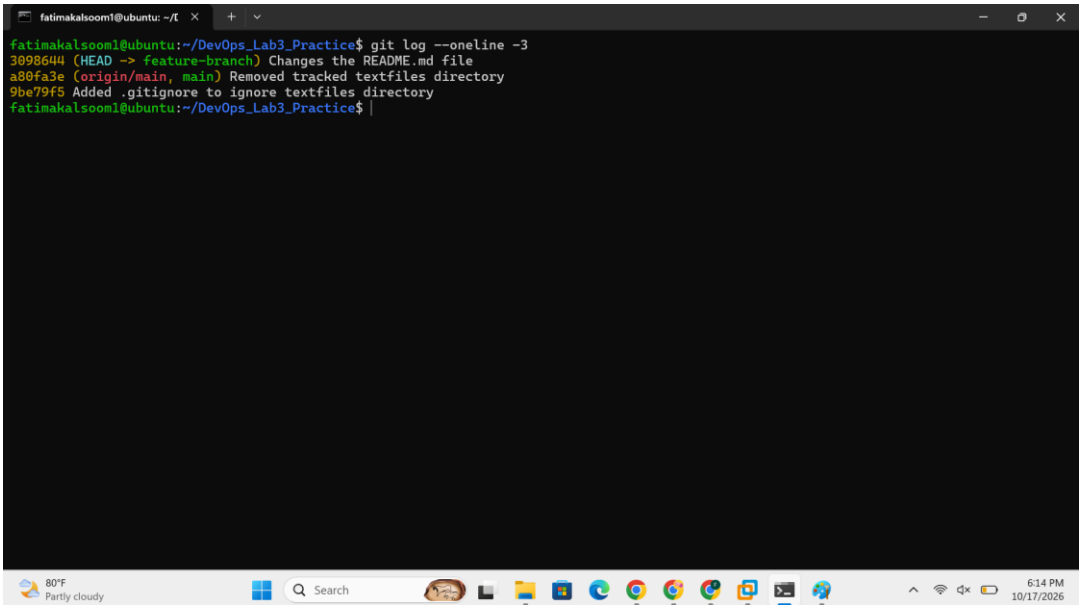


git rm -r --cached removes textfiles from tracking; change committed and pushed.

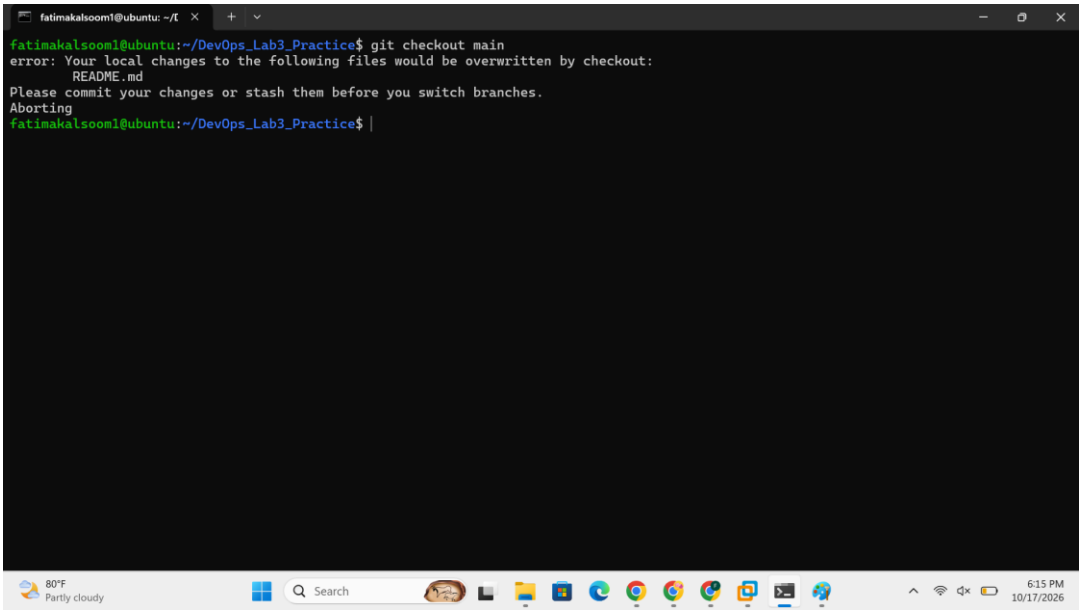


GitHub repository after the textfiles directory was removed from tracking.

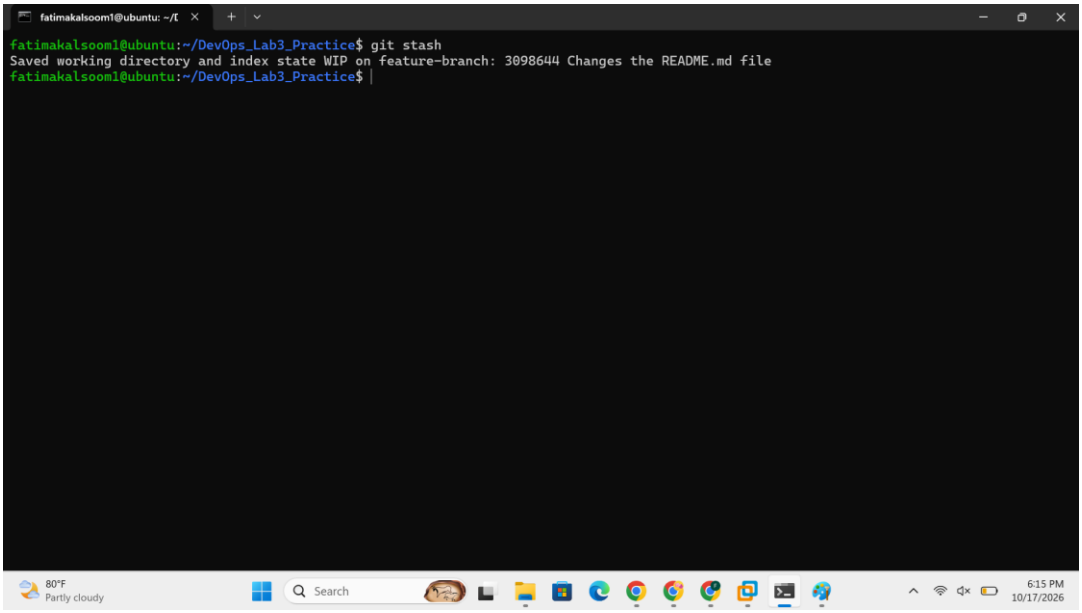
Task 4



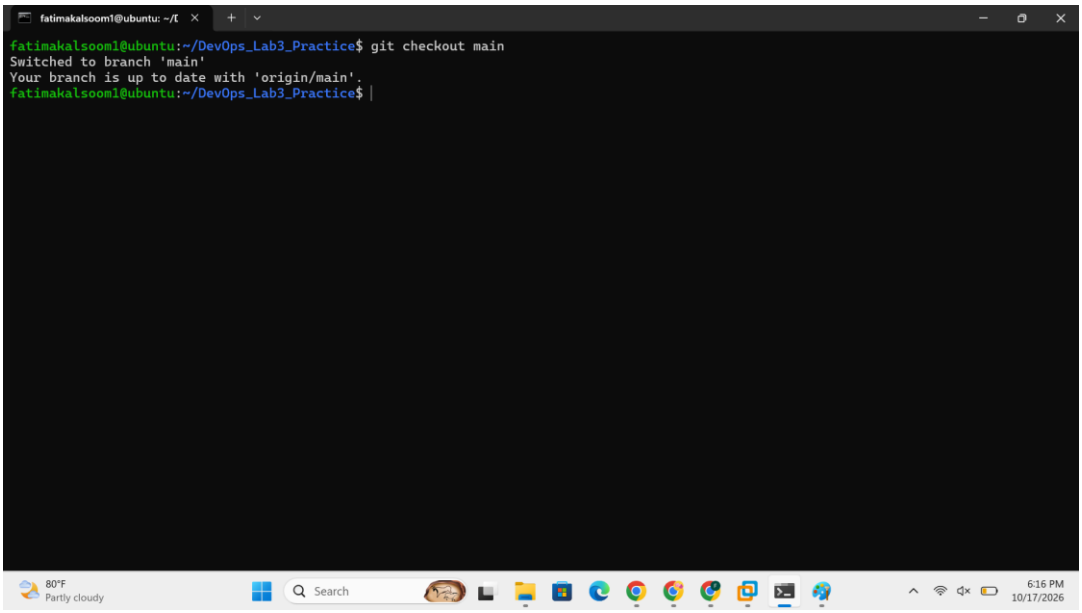
README.md modified and committed on feature-branch.



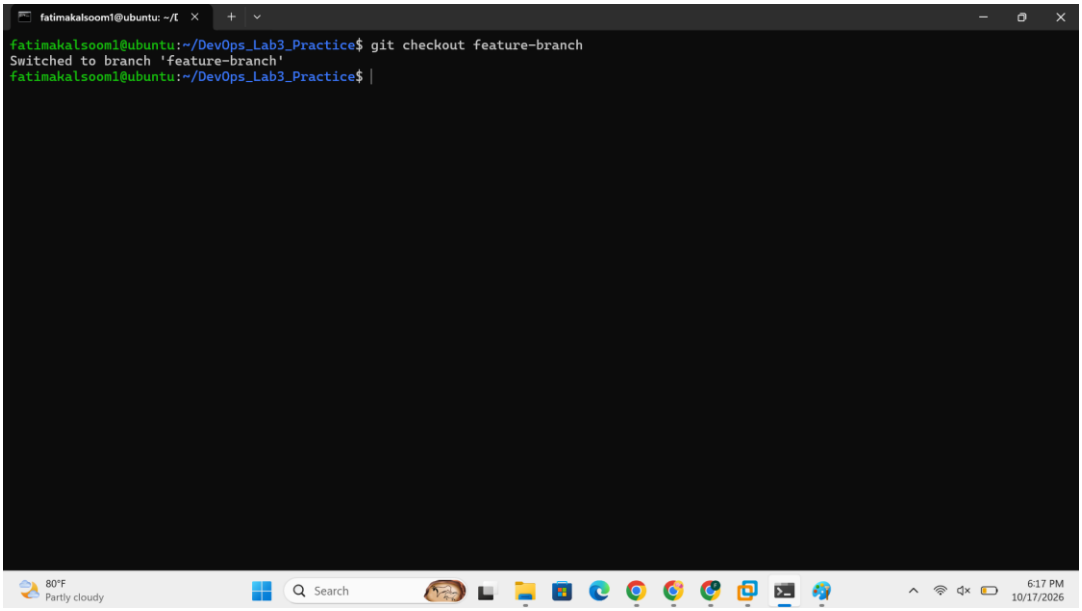
Checkout to main blocked by uncommitted local changes.



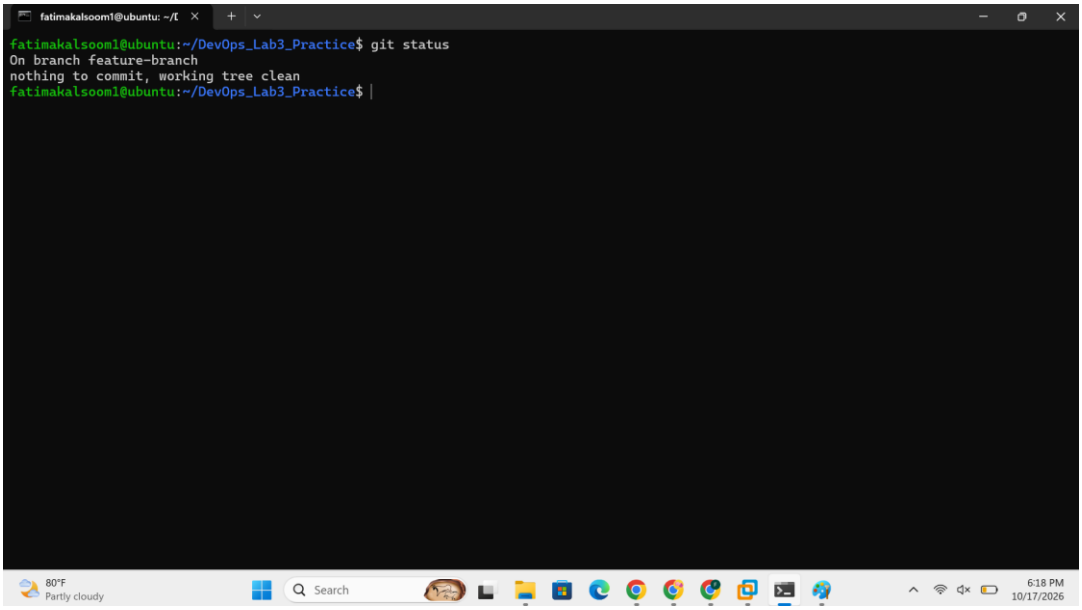
git stash saves the uncommitted changes.



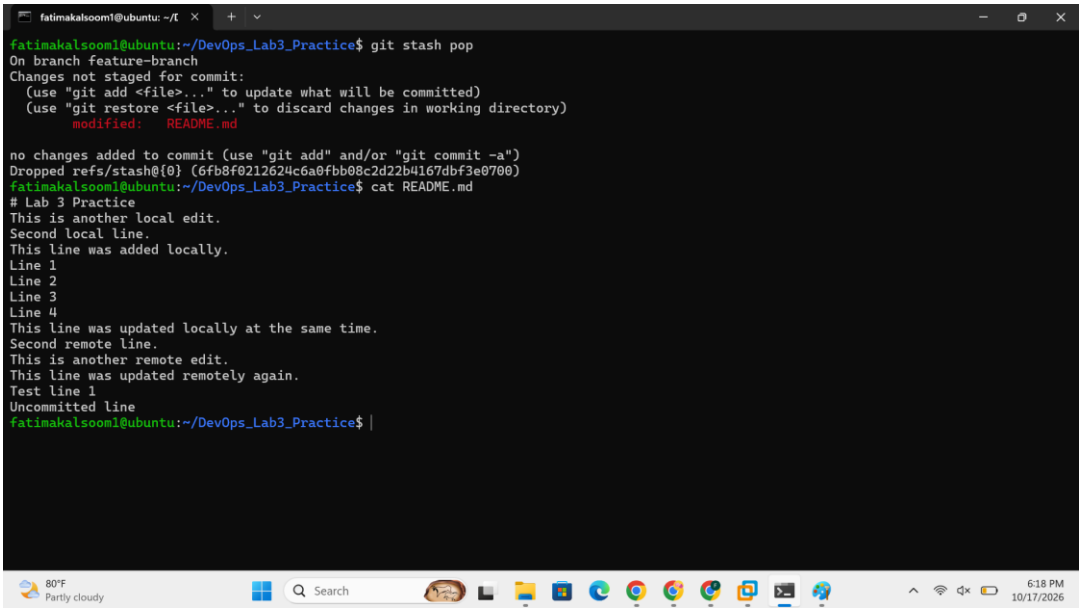
Switched to main after stashing.



Returned to feature-branch.

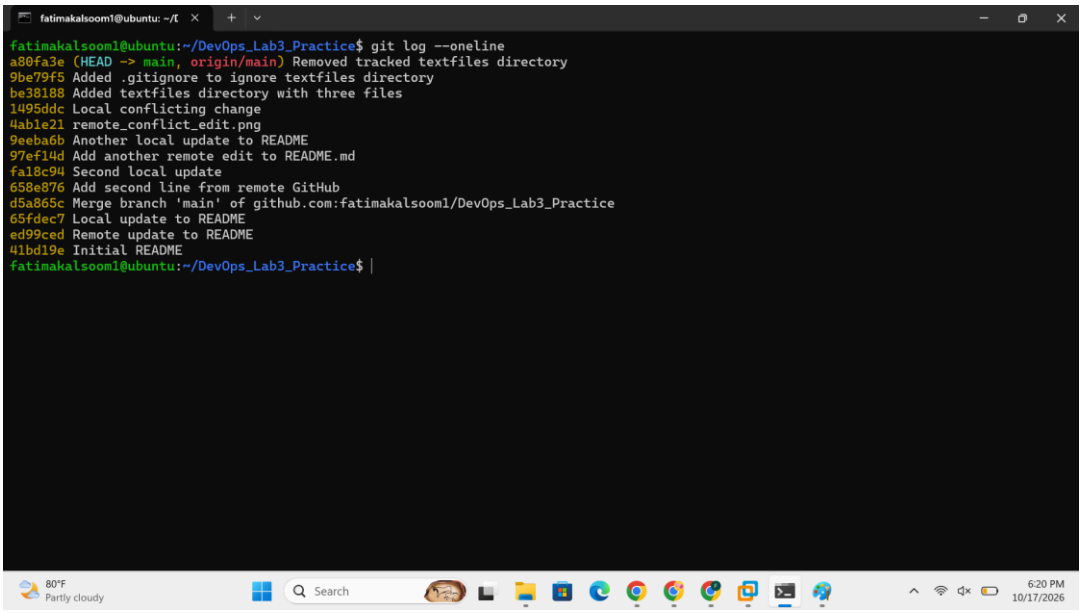


Working tree clean on feature-branch after stashing.

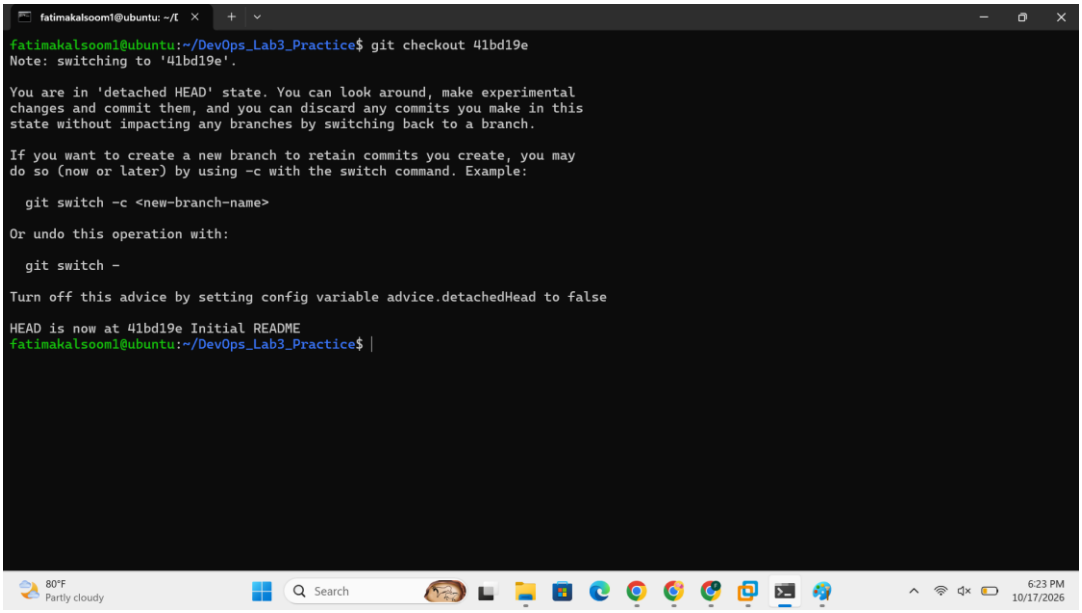


git stash pop restores the stashed changes.

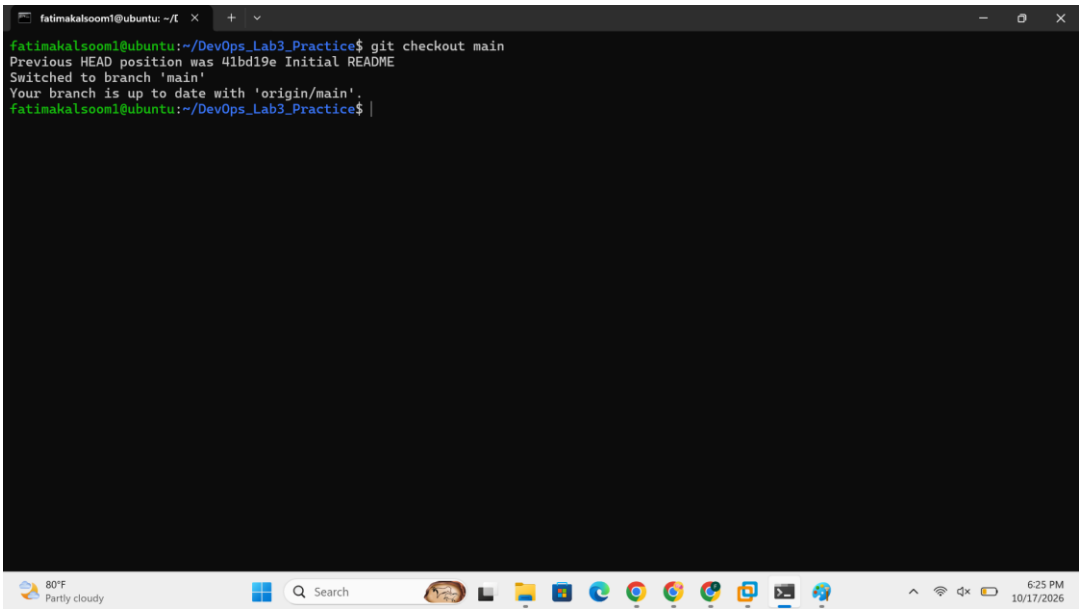
Task 5



Commit history shown with git log --oneline.

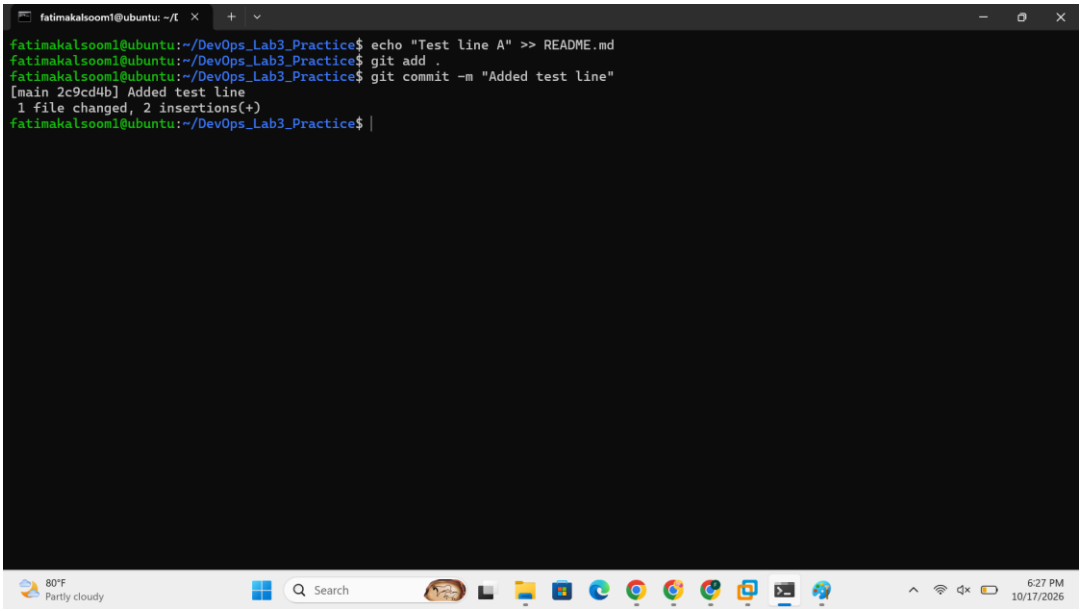


Checking out a commit hash puts the repository in detached HEAD state.

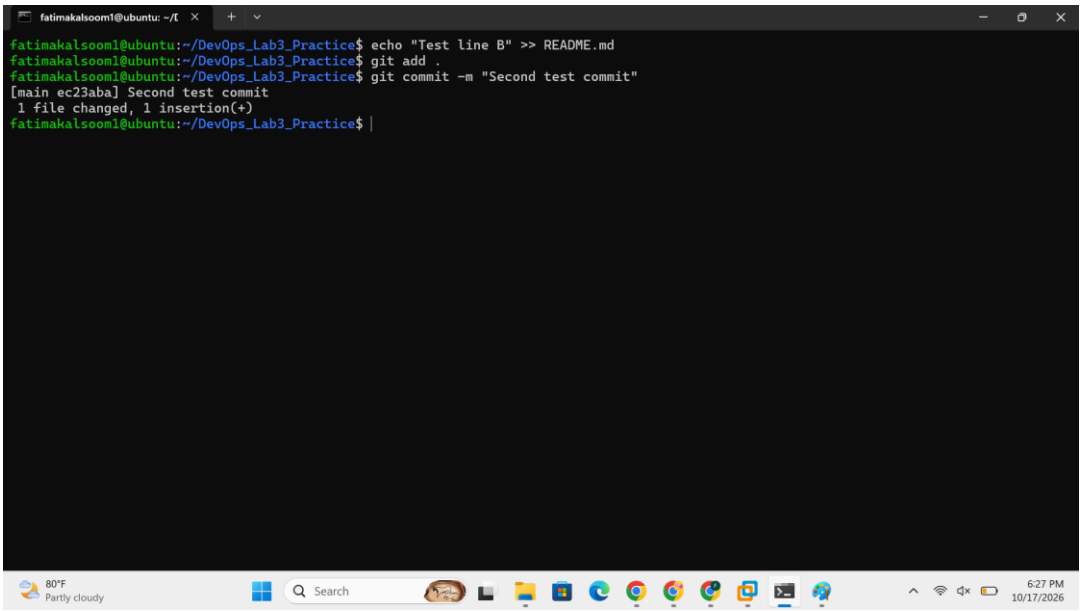


Returned to the main branch from detached HEAD.

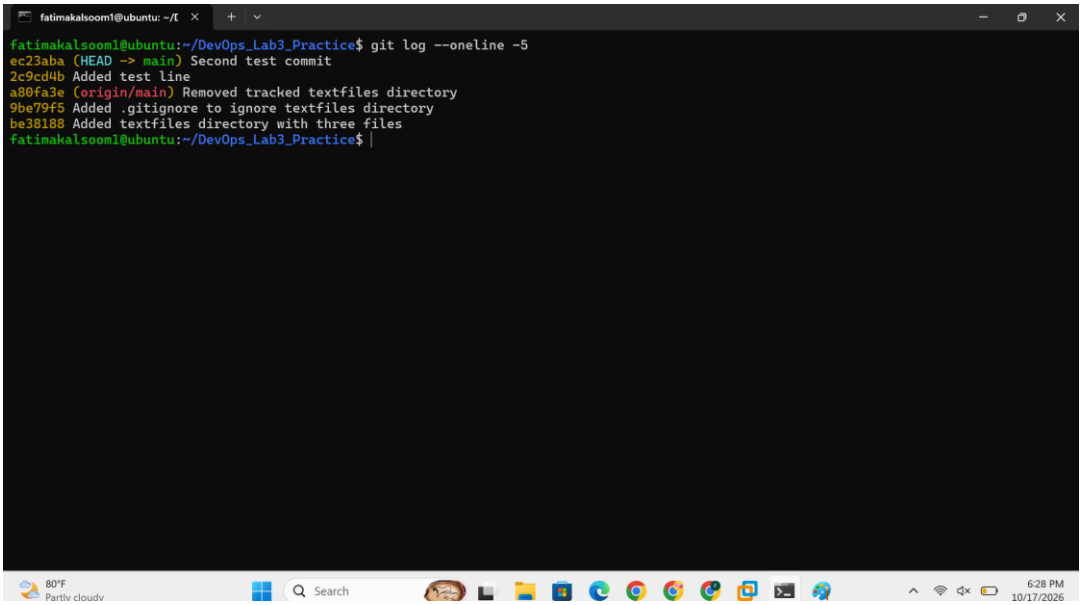
Task 6



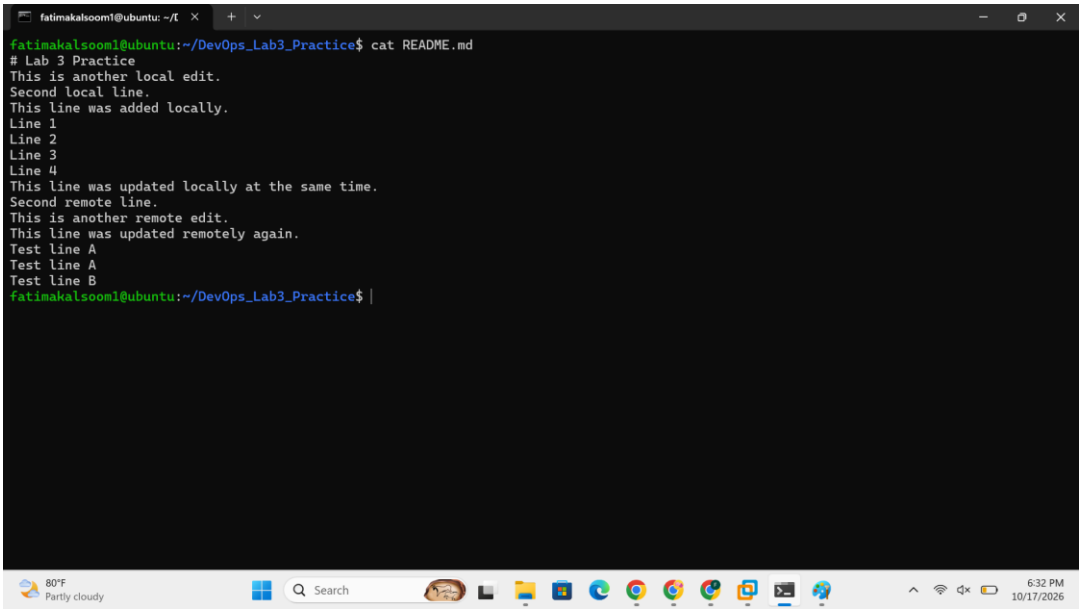
First test line added and committed.



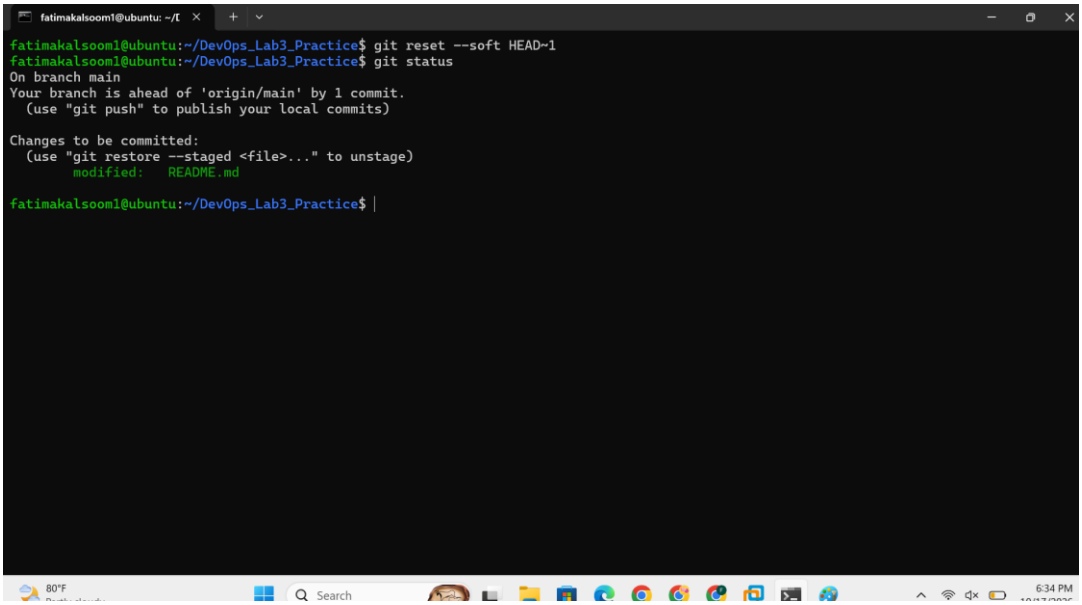
Second test commit created.



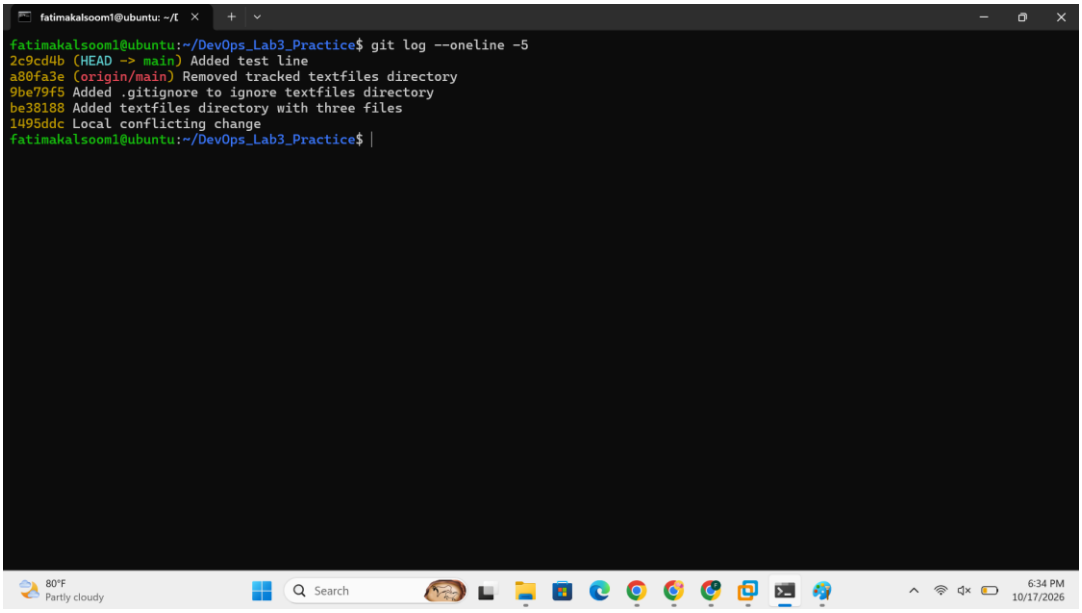
Commit history before the reset.



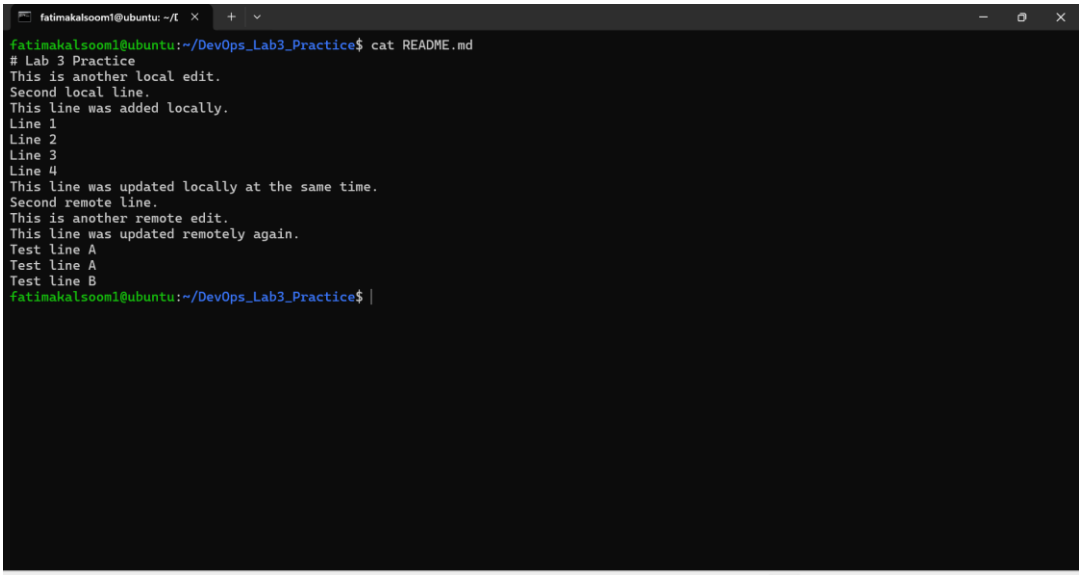
README.md contents before the reset, with both edits present.



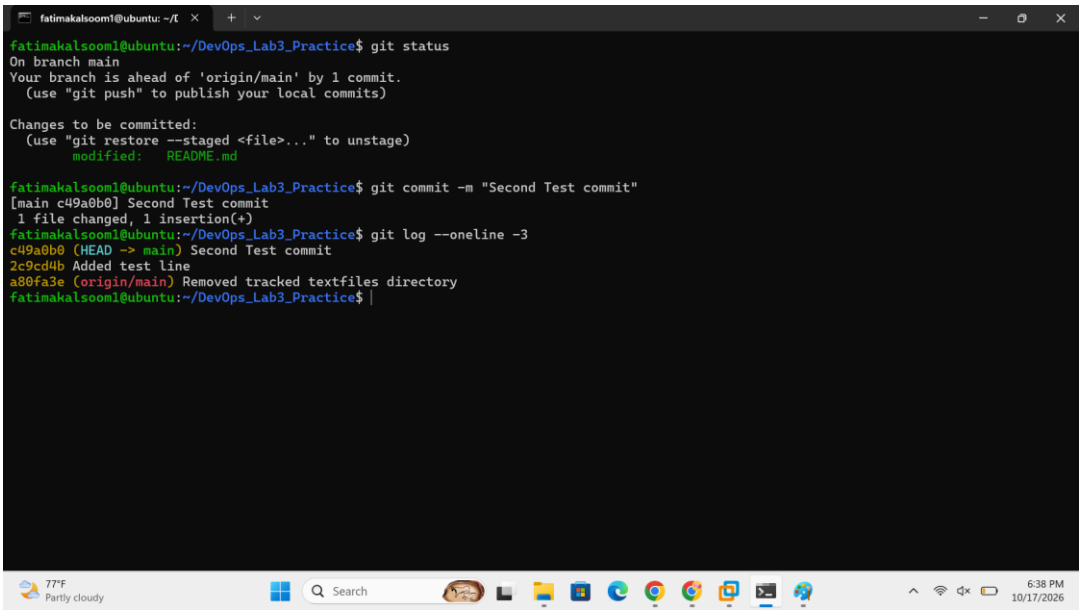
git reset --soft HEAD~1 keeps the changes staged.



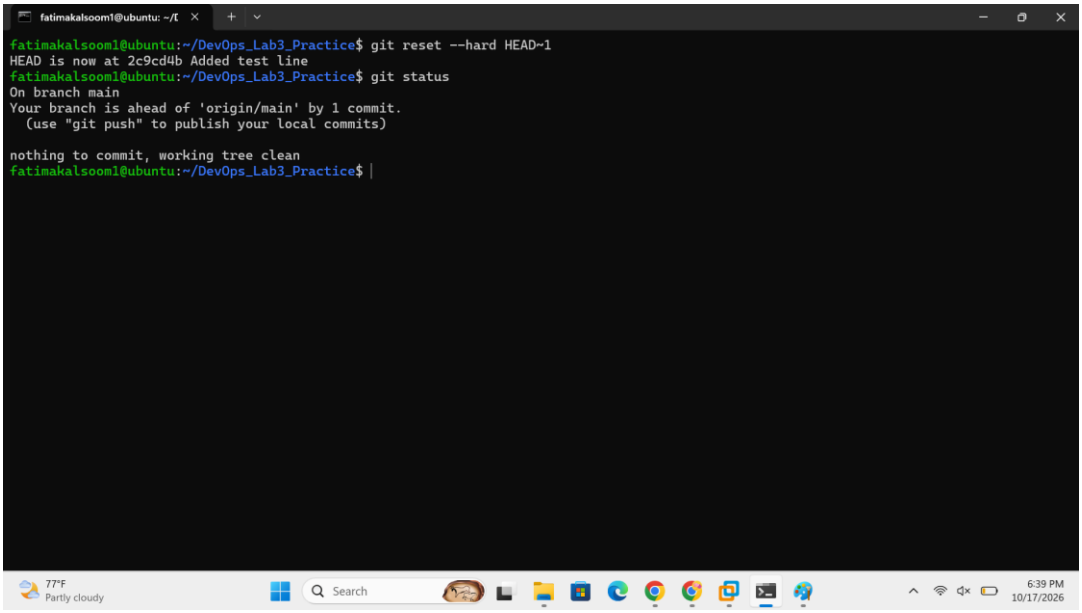
Commit history after the soft reset.



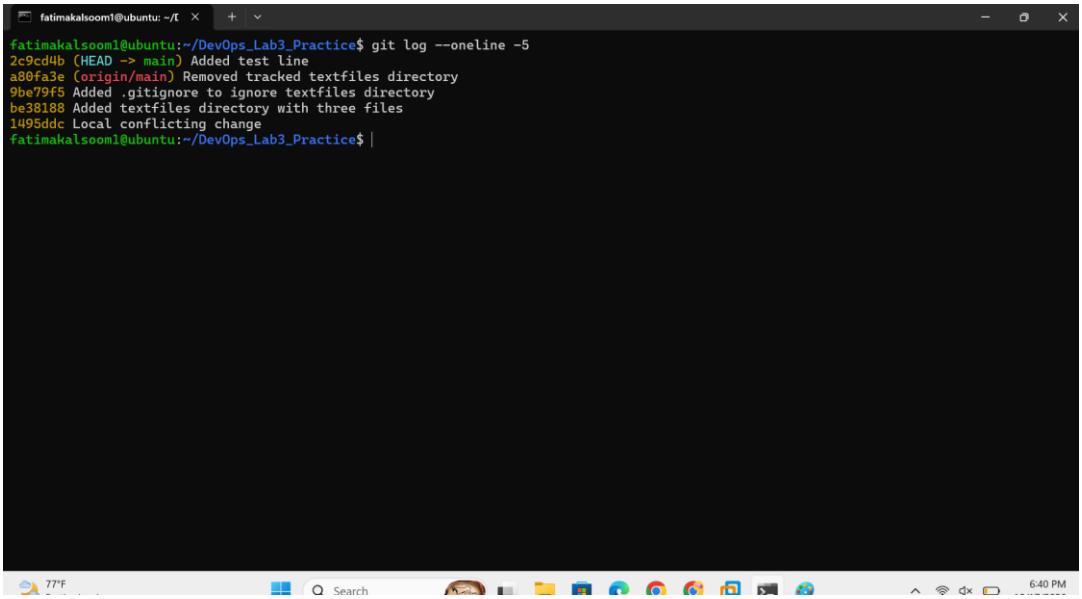
README.md still contains both edits after the soft reset.



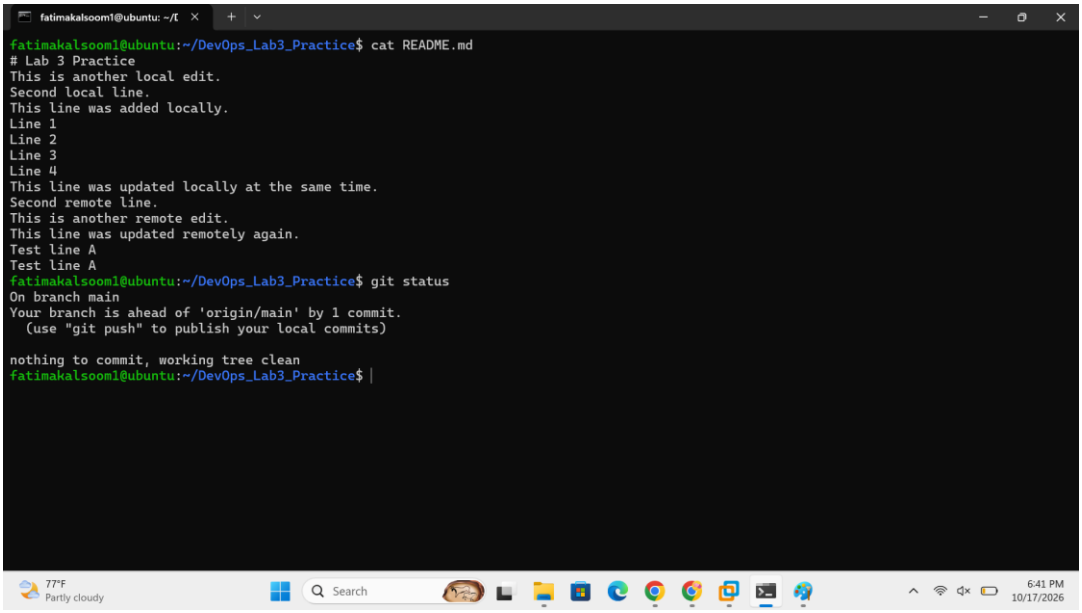
Staged changes recommitted after the soft reset.



git reset --hard HEAD~1 discards the last commit and its changes.

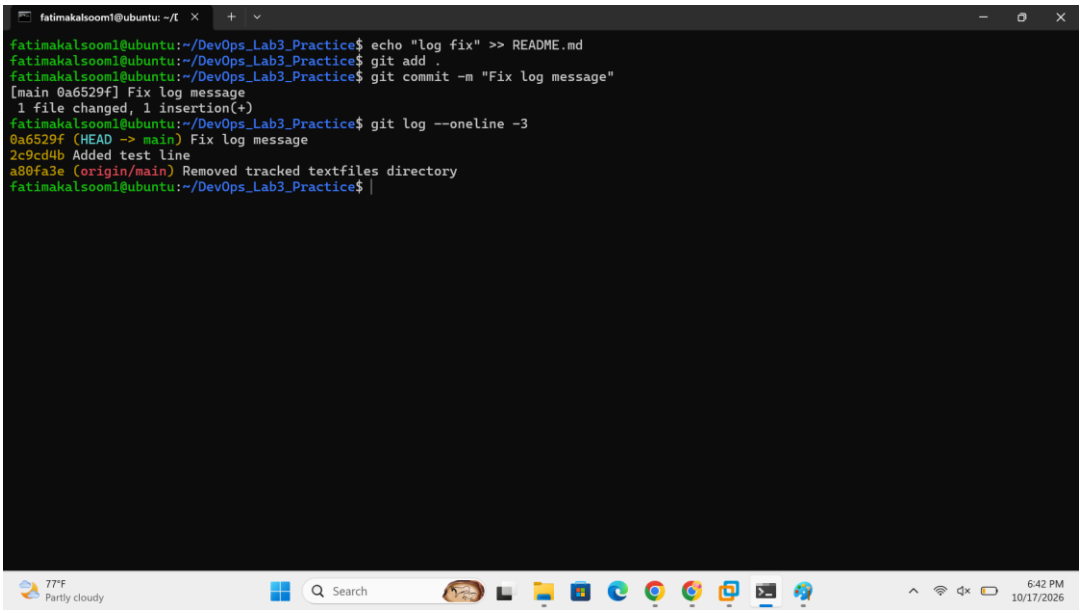


Commit history after the hard reset.

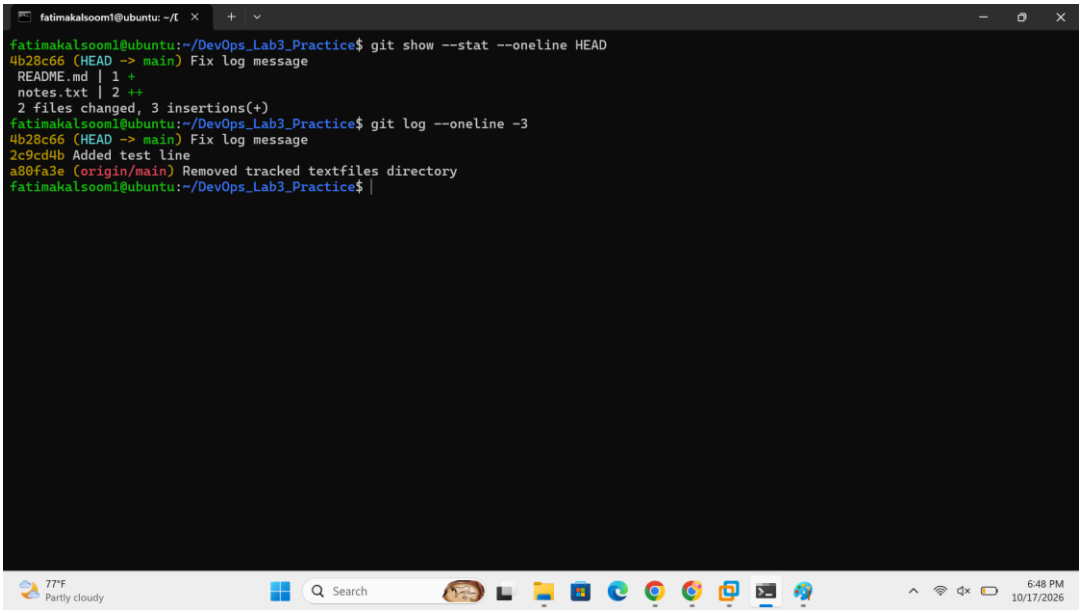


README.md after the hard reset; the second edit is gone.

Task 7

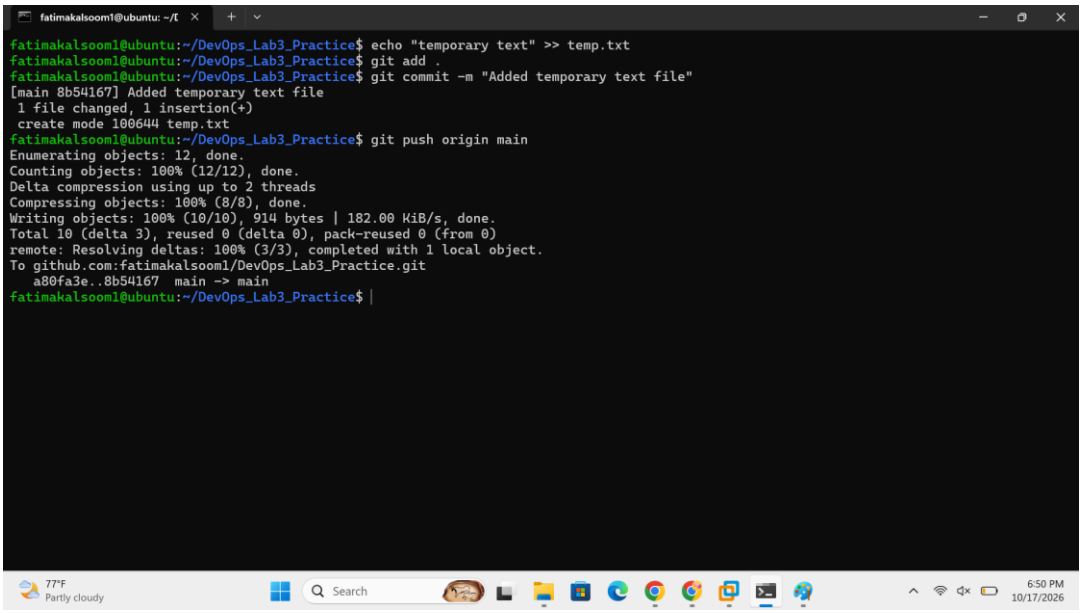


Commit "Fix log message" created.

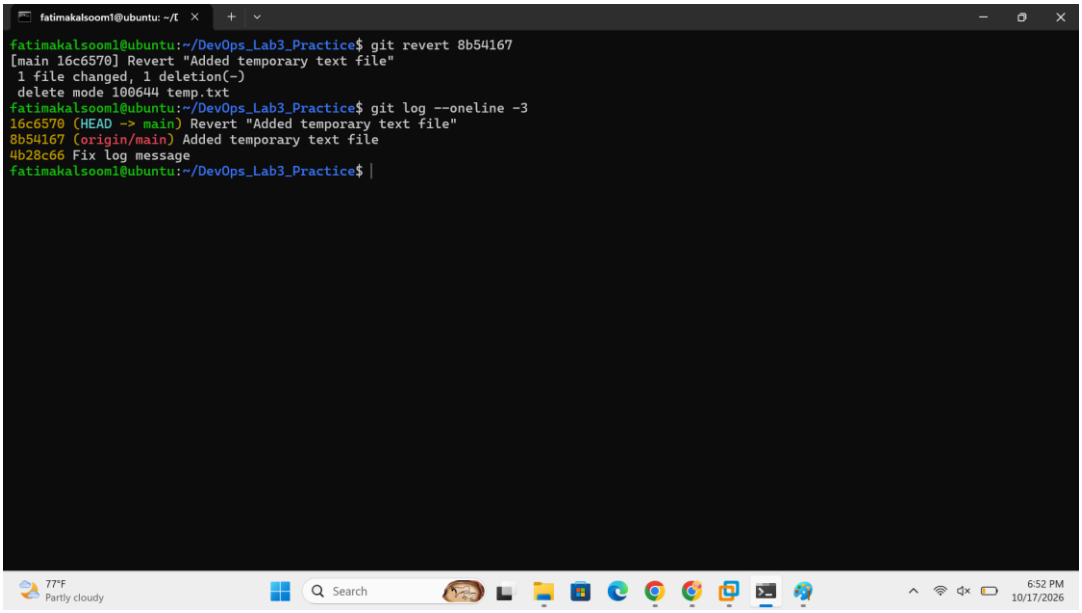


Last commit amended with git commit --amend; git show --stat confirms the change.

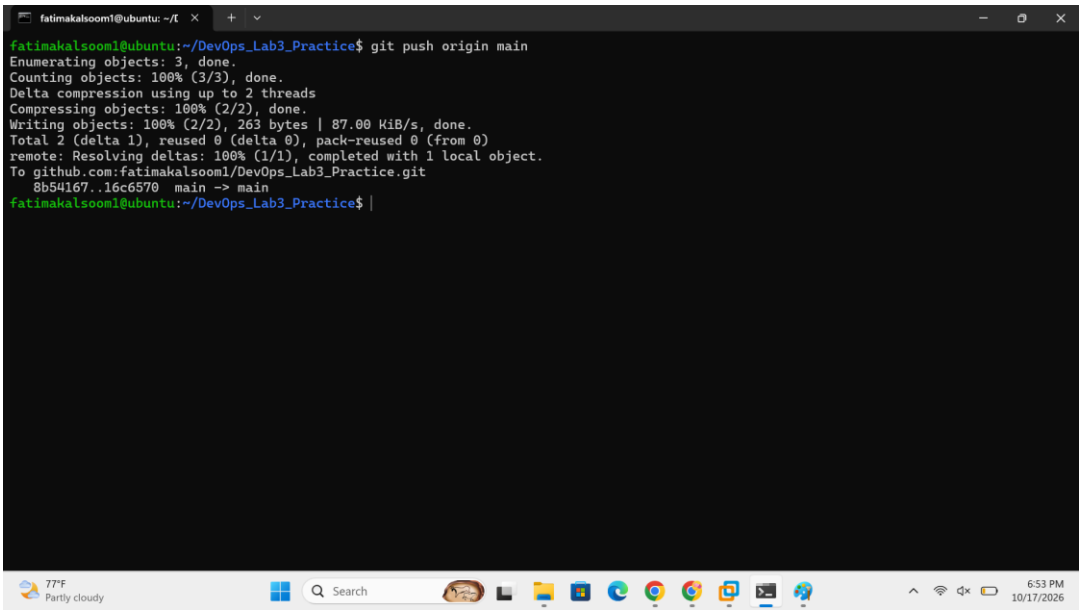
Task 8



Temporary file committed and pushed to origin/main.

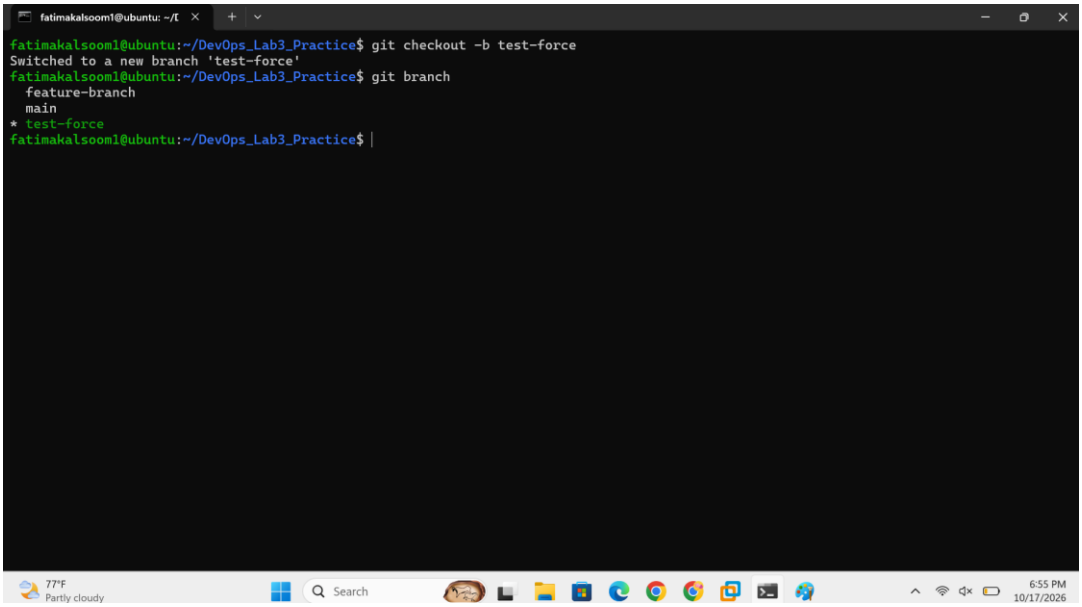


git revert creates a new commit that undoes the temporary file commit.

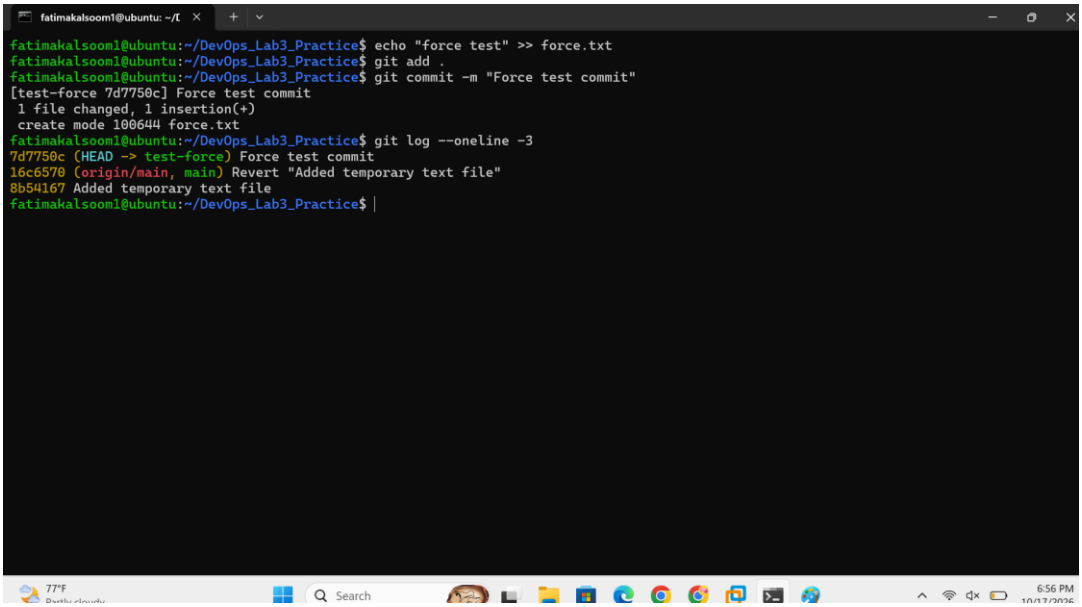


Revert commit pushed to origin/main.

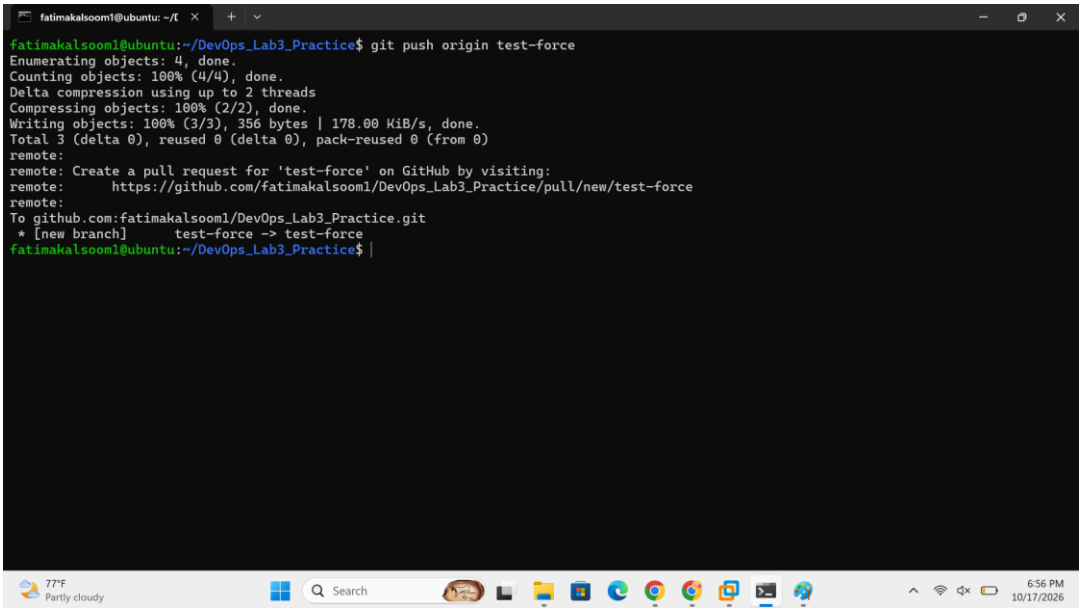
Task 9



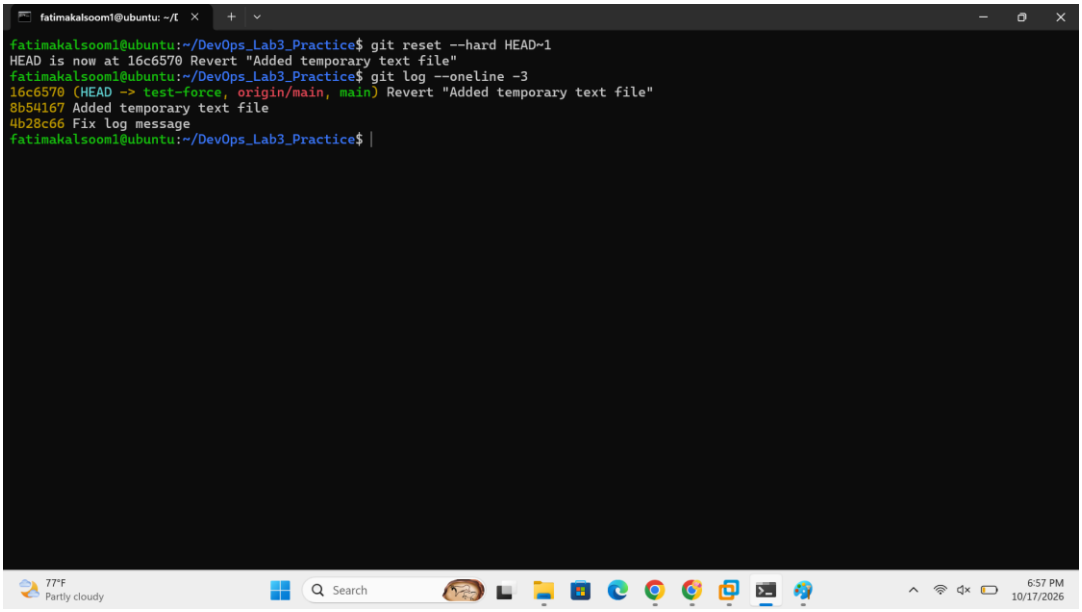
New branch test-force created.



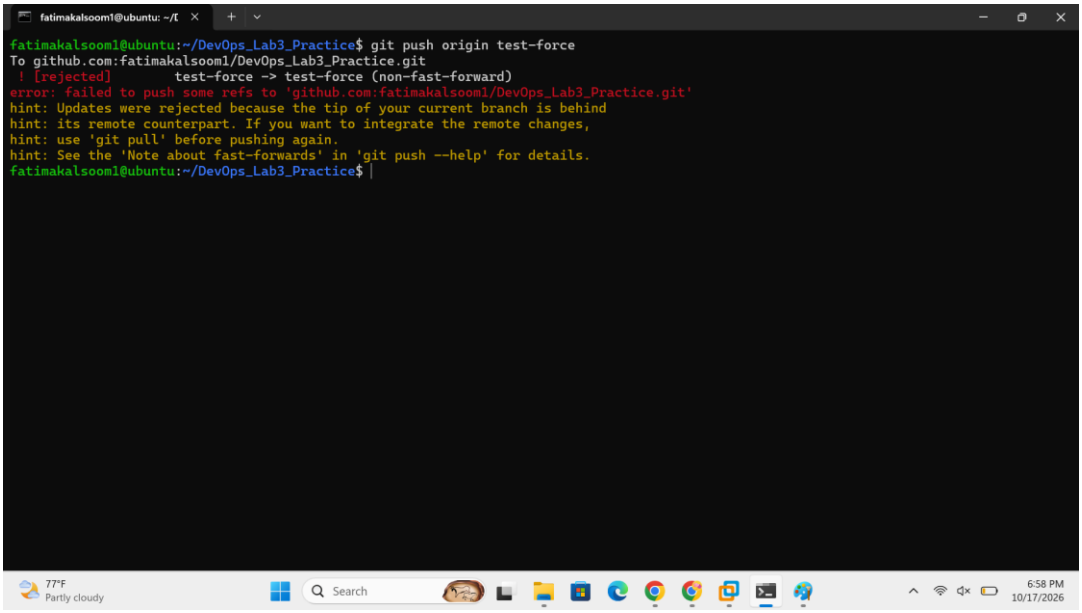
Commit made on test-force.



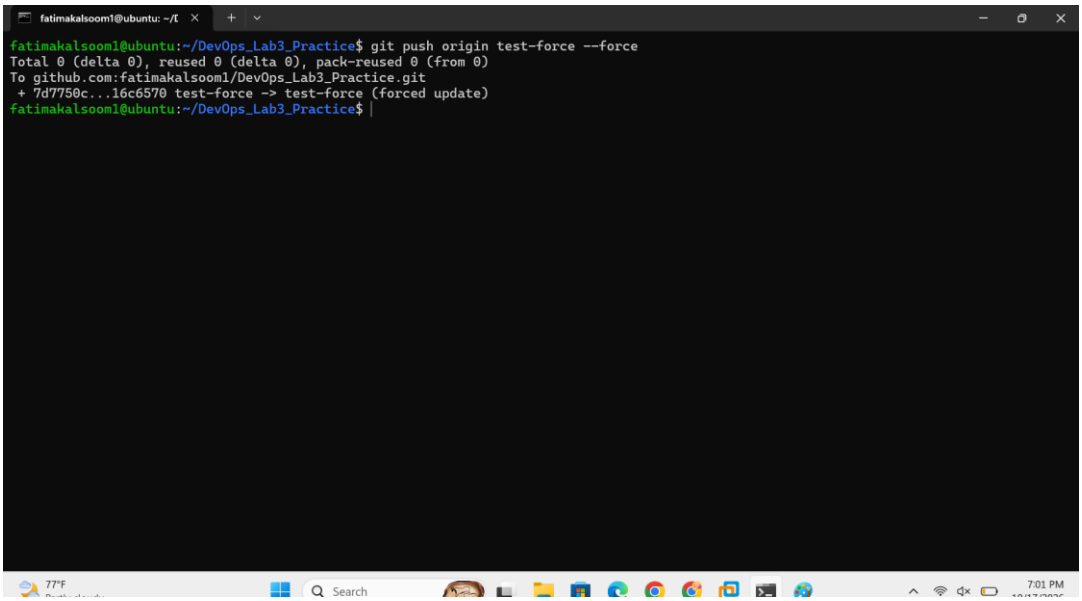
test-force branch pushed to the remote.



Local history rewritten with git reset --hard HEAD~1.

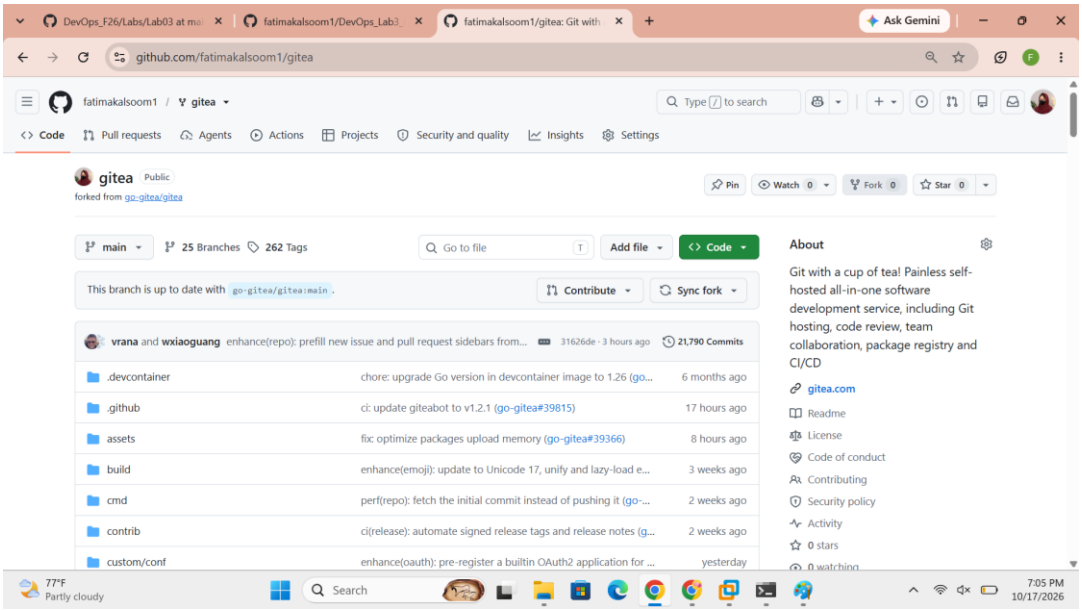


Normal push rejected as non-fast-forward after the reset.

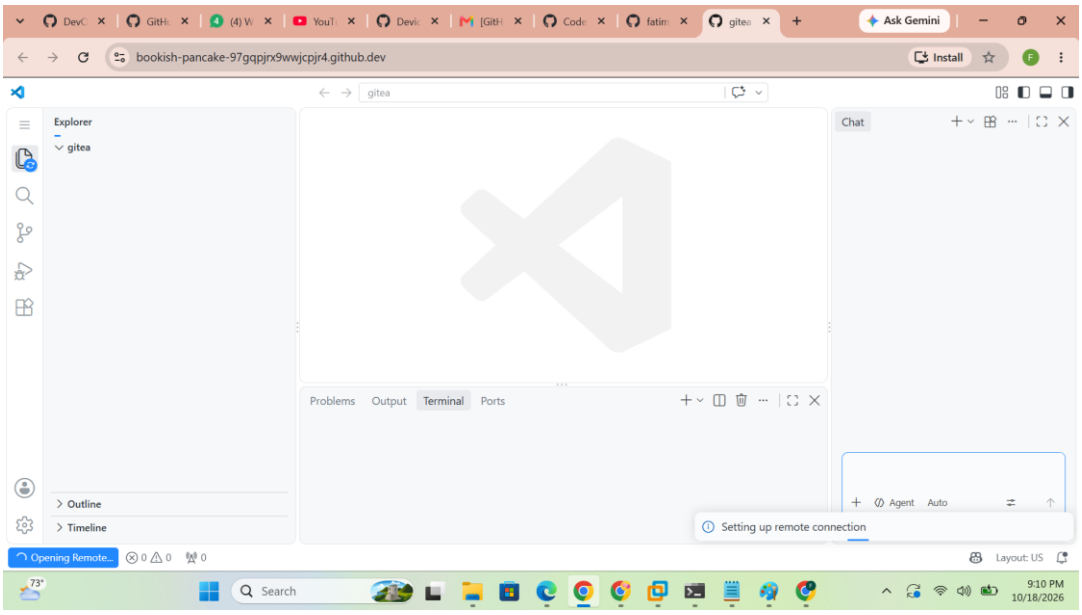


git push --force overwrites the remote branch history.

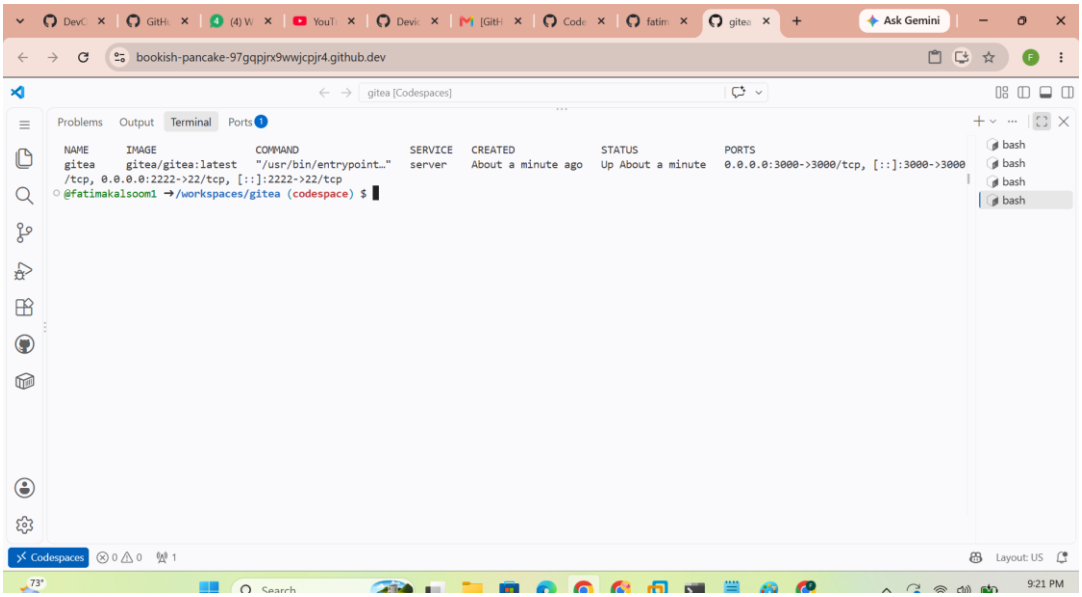
Task 10



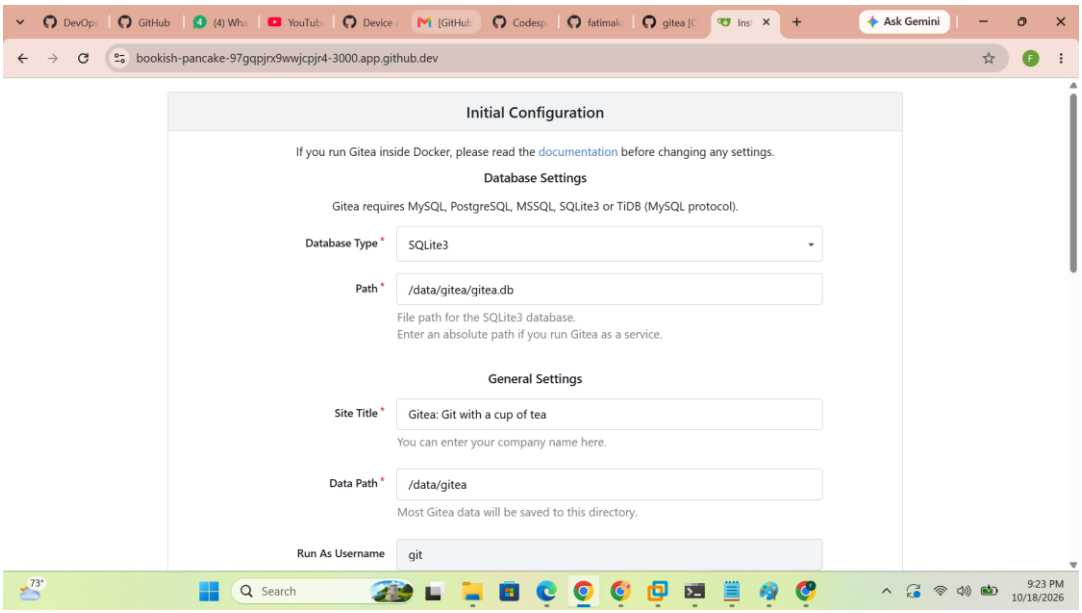
Gitea repository forked into my GitHub account.



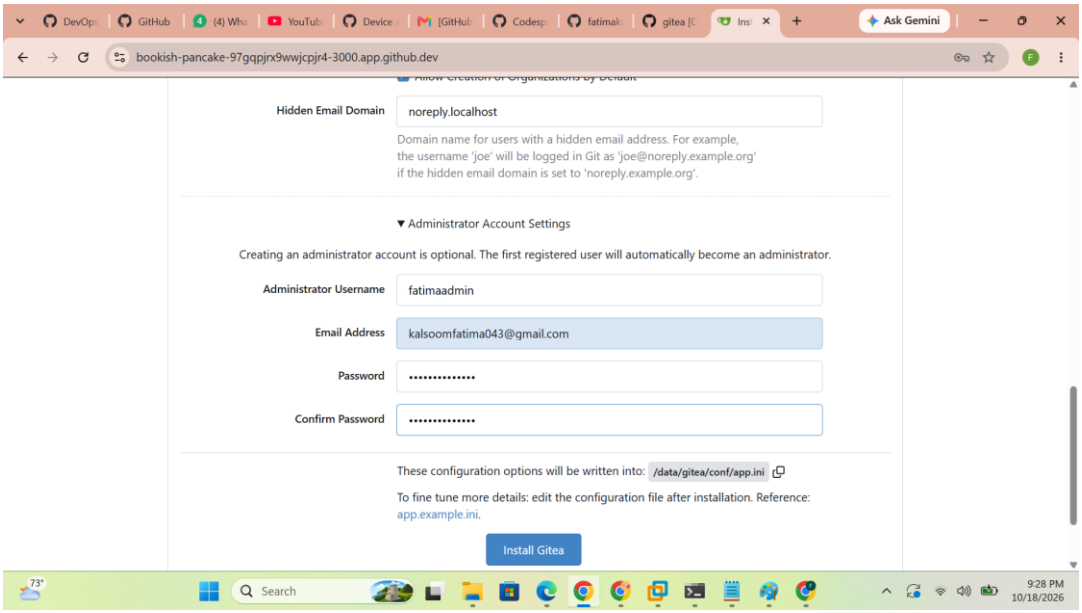
GitHub Codespace loading for the forked repository.



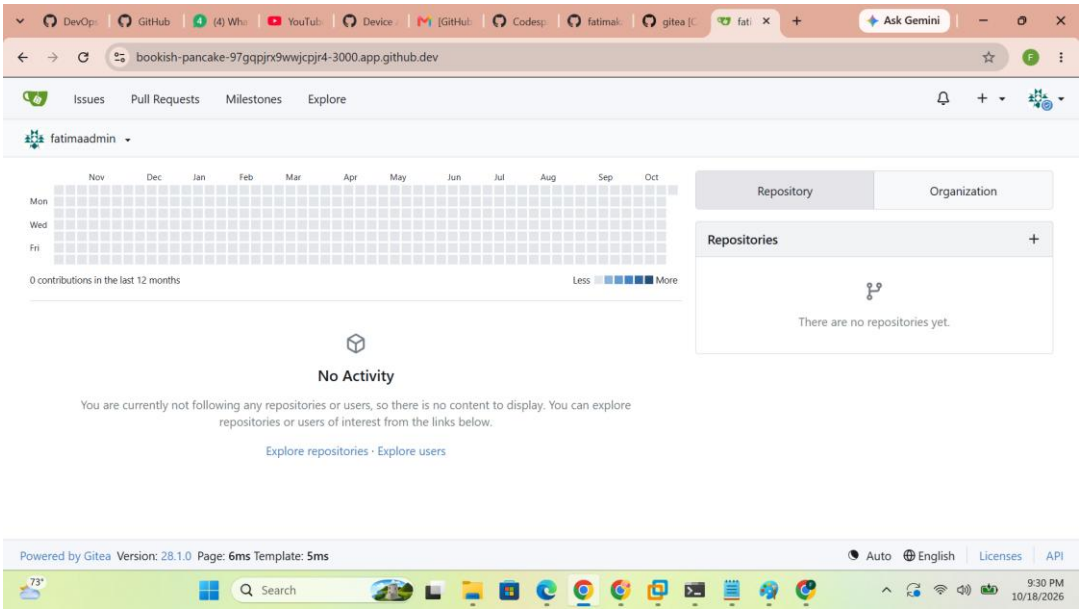
Gitea container running via docker compose up -d in the Codespace.



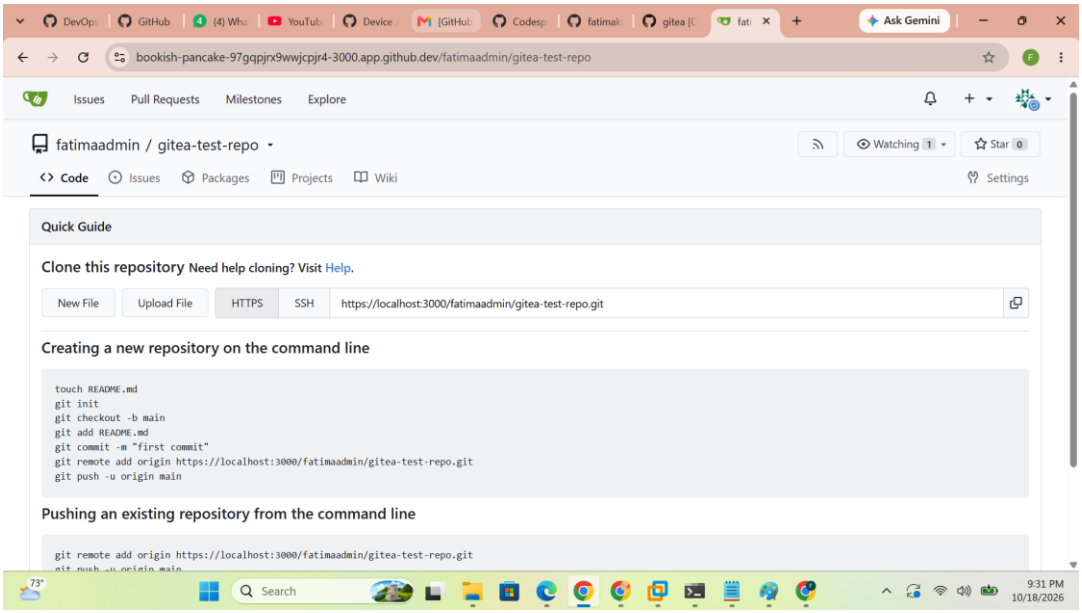
Gitea initial configuration page.



Administrator account settings filled in before installing.

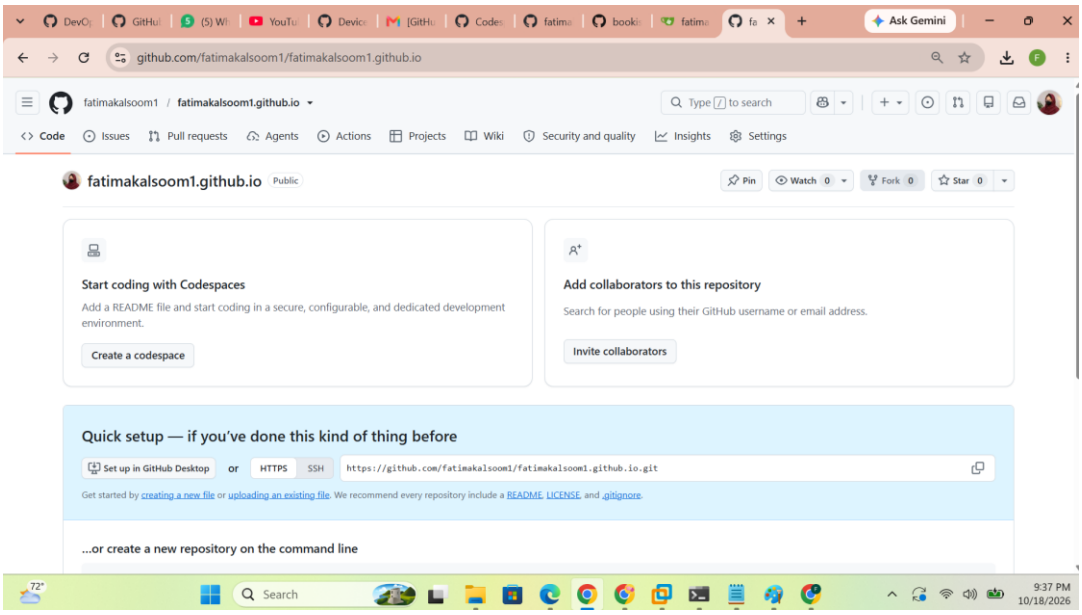


Gitea dashboard after login.

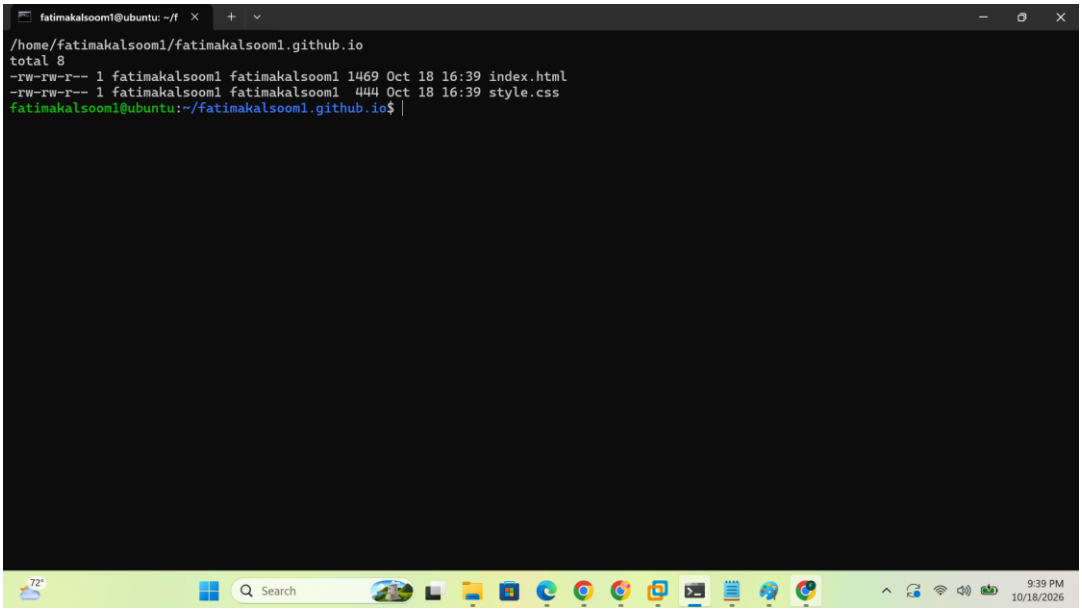


New repository gitea-test-repo created in Gitea.

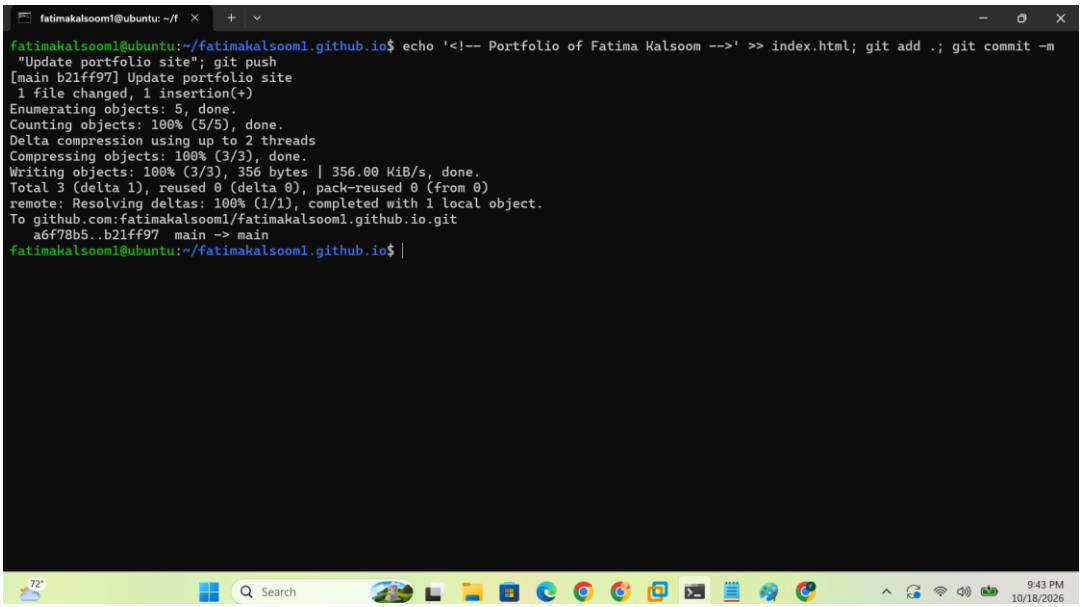
Task 11



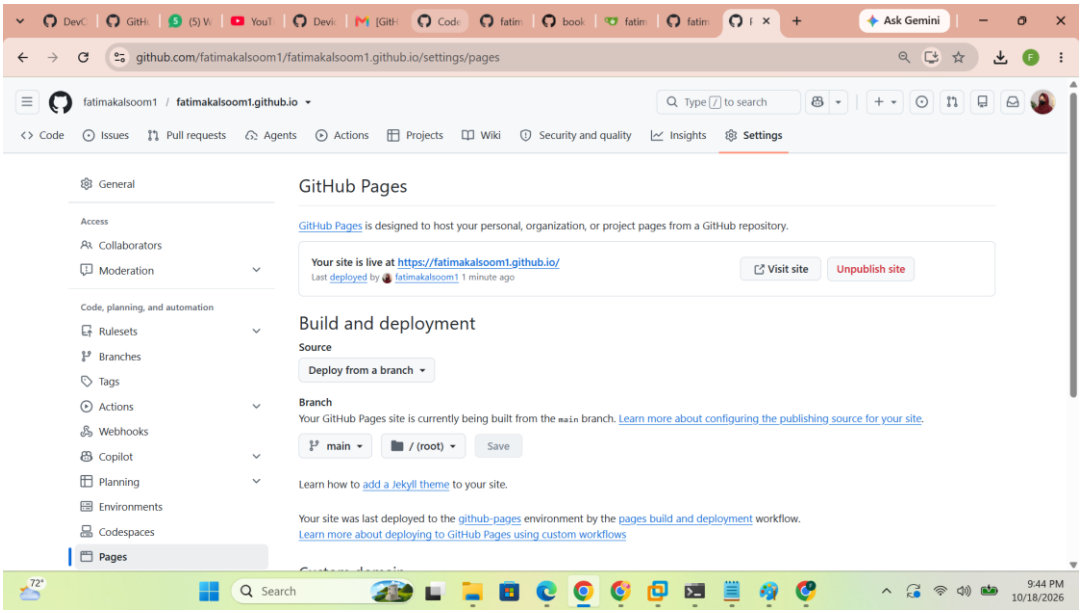
GitHub Pages repository fatimakalsoom1.github.io created.



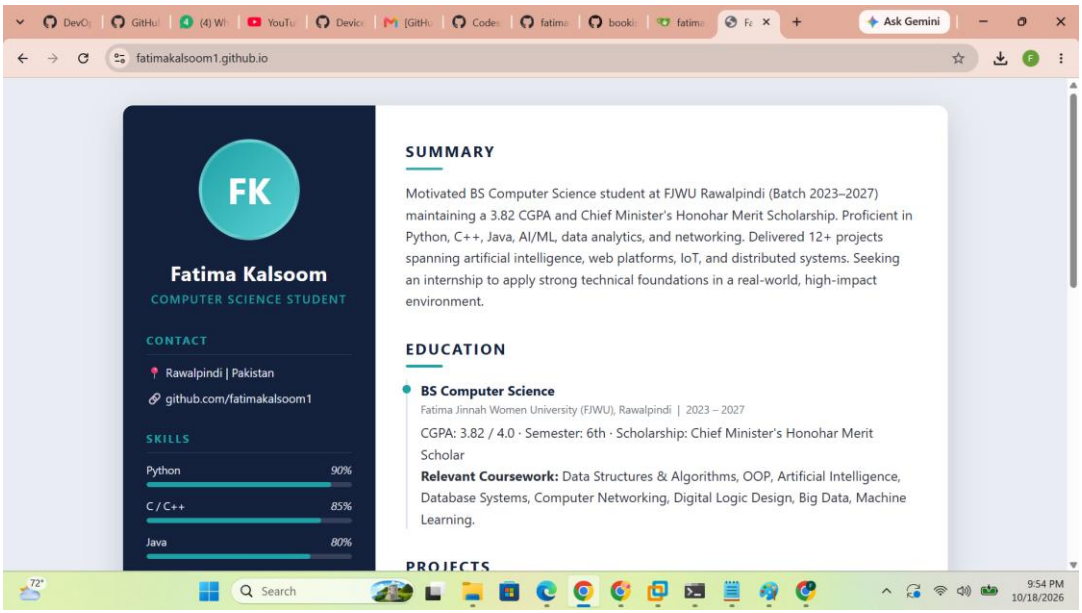
Static site files (index.html, style.css) in the local repository.



Site files committed and pushed to GitHub.



GitHub Pages settings showing the site is live.



Published portfolio site open in the browser.